HADAMARD DESIGNS AND INCIDENCE REPAIR FOR KAPLANSKY'S DIRECT-FINITENESS PROBLEM

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Abstract. We give a finite incidence construction for one-sided inverses in group algebras of torsion-free groups in every odd prime characteristic. Projective lines supply the required intersection congruences; Hadamard blocks provide regular auxiliary incidences; and unions of concurrent lines correct inverse-letter balances exactly over the integers. The repair-only union rows have fixed multiplicities, leaving a replicated bulk of lines and Hadamard-decorated lines whose reduced-turn kernel admits a uniform spectral bound. If $p \geq 5$ and a real Hadamard matrix of order $4p$ exists, a plane of order p^3 gives $p^6 + p^3 + 4p$ signed letters; Du–Jiang's existence theorem gives at most $p^6 + p^3 + 2^{10}p^{23/20}$ for $p \geq 2^{23}$. In characteristic three we specify a 6654-letter input with root degree 93 and maximum graph degree 325. The signed-matching and bounded-path methods of OpenAI and Ivanov, together with Mghirbi–Douglas's cone criterion, realize these data in finitely presented torsion-free word-hyperbolic groups, with a finite classifying complex of dimension at most two and witnesses $ab = 1$, $ac = 0$, $c \neq 0$. The contribution is a quantitative refinement of the finite input to the qualitative group-algebra result of Mghirbi–Douglas.

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1. Introduction

A unital ring is *directly finite* if $ab = 1$ forces $ba = 1$. Direct finiteness of group algebras holds in characteristic zero [25], and is known for many classes of groups in arbitrary characteristic, including free-by-amenable and sofic groups [1, 12]. Kaplansky's direct-finiteness problem is related to, but distinct from, his zero-divisor and idempotent conjectures for torsion-free groups [17]. A zero divisor alone need not give a one-sided inverse. Conversely, if $ab = 1$ but $ba \neq 1$, then $e = ba$ is an idempotent distinct from 0 and 1: indeed $e^2 = e$, and $e = 0$ would force $a = a(ba) = 0$. Thus a failure of direct finiteness also gives a counterexample to the idempotent conjecture.

Mghirbi and Douglas [23, Theorem C] recently constructed, in every odd prime characteristic p, a finitely presented torsion-free word-hyperbolic group G with a finite two-dimensional classifying space and elements $a, b, c \in \mathbb{F}_p[G]$ such that

$$ab = 1, \qquad ac = 0, \qquad c \neq 0. \tag{1.1}$$

Their realization builds on labelled-graph and cancellation methods of OpenAI and Ivanov. The present paper changes the *finite incidence input* to that method. Its purpose is quantitative: the displayed Mghirbi–Douglas direct-finiteness construction has a signed alphabet of order p^{10}, whereas the constructions here achieve eventual order p^6. In characteristic three the signed alphabet decreases from 59306 to 6654 letters, though the maximum graph degree rises from 257 to 325. These are comparisons of the specified constructions, not claims about minimum possible alphabets.

The finite construction is illuminated by the characteristic-three case, where projective-plane incidences provide explicit cancellation data. Extending the design across odd prime characteristics makes two arithmetic requirements prominent: a pairing of inverse-letter residues and exact integer balances, not merely congruences. Two affine pencils address the pairing, while Hadamard blocks provide auxiliary intersections with uniform replication statistics. The balance corrections pose a further difficulty: unions of concurrent lines repair integer coordinates, but need not have suitable leading turn weights. We therefore replicate only lines and Hadamard-decorated lines, retaining the union rows at fixed multiplicities. Exact repair then leaves the contracting leading turn distribution undisturbed. The finite input is explicit up to the terminal matching, obtained by a terminating existence-based search.

1.1. Results.

A signed alphabet consists of oriented labels paired by a fixed-point-free involution; each inverse pair provides a generator of the initial rose. Definition 2.1 specifies the finite matching data, including incidence congruences, exact slot balances, and a strict reduced-turn estimate.

Theorem 1.1 (Quantitative finite constructions)**.** *Finite input in the sense of Definition 2.1 exists in every odd prime characteristic. The following choices are available.*

(i) *In characteristic three, the signed alphabet has* 6654 *letters, the exceptional root has degree* 93*, and the maximum matching-graph degree is* 325*. The primitive leading bulk is* 132388347*, with a fixed reservoir of* 39858 *vertices in each class.*

(ii) *If $p \geq 5$ and a real Hadamard matrix of order $4p$ exists, the ordinary plane has order p^3 and the signed alphabet has $p^6 + p^3 + 4p$ letters.*

(iii) *For every prime $p \geq 2^{60}$, the parameter choice from Ghaderpour's stated Hadamard bound gives plane order p^3 and at most $p^6 + p^3 + 2^{15}p^{6/5}$ signed letters.*

(iv) *For every prime $p \geq 2^{23}$, the Hadamard-existence theorem of Du and Jiang gives plane order p^3 and at most $p^6 + p^3 + 2^{10}p^{23/20}$ signed letters.*

In (ii)–(iv)*, the maximum matching-graph degree is at most $(p + 1)p^3 + 1$. All finite data are fixed before the replication parameter varies.*

For the unconditional estimates in (iii) and (iv), the required Hadamard matrices follow from Du–Jiang [10, Theorem 3], using Sylvester doubling to obtain the orders associated with Ghaderpour's parameter choice [14, Theorem 10]. Part (ii) assumes an actual matrix of order $4p$ and does not depend on the Hadamard conjecture. The general construction appears in

Theorem 4.1, the quantitative specializations in Section 5, and the complete characteristic-three input in Section 6.

Corollary 1.2 (Group-algebra realization)**.** *Each finite construction in Theorem 1.1 admits a realization by a finitely presented torsion-free word-hyperbolic group G, with a finite classifying complex of dimension at most two, satisfying* (1.1)*. It has a presentation on $|T|/2$ alphabet generators. The same witnesses work over every field of characteristic p; in particular $ba \neq 1$ and ba is an idempotent different from 0 and 1.*

The realization is Theorem 8.5. The alphabet bound is a bound on generators in the indicated presentation, not on the size of the matched graph or on collected group-algebra supports. The quantitative analysis keeps the finite input fixed while increasing the replication parameter.

1.2. **Incidence, balance, and realization.** Three requirements govern the construction, and none implies the other two. Intersection congruences among outgoing-label sets encode the algebraic cancellation. Exact *integer* balances between inverse-letter slots are needed to form the graph. Finally, a path estimate must allow that graph to be chosen so its cone complex has the required geometry. Congruences alone do not supply matchings, and exact balance alone does not supply the geometric realization.

The cancellation mechanism explains the first requirement. In a bipartite labelled component give vertices alternating signs $\sigma(x)$. If g_x, h_y are group elements represented by paths from roots in two components, simultaneous traversal of an edge with label t preserves $g_x h_y^{-1}$ while reversing $\sigma(x)$. Pairing traversals therefore gives the integer identity

$$\sum_{x,y} |S_x \cap S_y|\, \sigma(x) g_x h_y^{-1} = 0,$$

where S_x is the set of outgoing labels at x. This signed cancellation, from Ivanov [16, Lemma 2.2] and applied to equal components by Mghirbi–Douglas [23, Section 7], is proved for the present graphs in Lemma 8.4. Ordinary intersections congruent to 1 modulo p produce a zero product. A single exceptional self-intersection congruent to 0 gives the missing unit term and hence $ab = 1$; a second component gives $ac = 0$. Root protection in the final group supplies $c \neq 0$.

Projective planes provide the congruences with little algebraic overhead. Two distinct lines meet once, and a line in $\mathrm{PG}(2, q)$, for q a power of p, has $q + 1 \equiv 1 \pmod{p}$ points. Unions R_x of $p + 1$ concurrent lines retain the required intersection residues, while the integer identity

$$p\mathbf{e}_x = \sum_{j=1}^{p+1} \mathbf{1}_{L_{x+\ell_j}} - \mathbf{1}_{R_x}$$

permits a correction by a multiple of p at a single coordinate. The projective geometry thus provides both the ordinary rows and a mechanism for exact repair.

Auxiliary letters require a second incidence design. A normalized Hadamard matrix gives blocks whose intersections are multiples of p and whose point frequencies and pair frequencies are uniform. Its two Gram identities have complementary uses: one preserves the modular intersection conditions, and the other gives the one-letter pools and the conditional turn weights. These weights, rather than the joint frequencies alone, determine the squared-turn contraction. Sections 2 and 4 make this distinction precise.

Choosing which ordinary and auxiliary letters are inverses is itself an arithmetic problem. A suitable sum w of ordinary line rows must satisfy $w_t + w_{\iota(t)} \equiv 1 \pmod{p}$. Two affine pencils allow us to arrange the required residue counts: at each affine point their coefficients add, so the histogram is a cyclic convolution. Their centres and common line at infinity supply controlled corrections. Once the involution is fixed, the line–union identity lifts the congruence to the exact equality

$$X + \iota Y = \mathbf{1}_{L_0 \cup E}.$$

This is the content of Section 3; the exceptional row $L_0 \cup E$ is the compact root already familiar from the Mghirbi–Douglas construction.

For the probabilistic estimates, exact repair must leave the leading turn statistics intact. The leading population uses only lines and Hadamard-decorated lines. A balanced finite reservoir retains the union rows, and the integer repairs enter as fixed affine offsets. All errors in two-coordinate candidate counts are therefore bounded independently of replication. Since the leading letter pools are positive and the squared reduced-turn kernel is strictly contracting, a small fixed inflation dominates the complete finite populations once replication is large. The resulting uniform estimate is the bridge to conditioned matchings and bounded-path exclusion in Section 7 and Appendix A. Mghirbi–Douglas's cone criterion then provides asphericity, root protection, and hyperbolicity, as described in Section 8 and Appendix B.

1.3. **Relation to previous work.** The group-theoretic realization is not claimed as a new general method here. OpenAI's direct-finiteness construction [28], building on its zero-divisor construction [29], develops the exceptional-root assembly, conditioned switching, bounded-path and cone-picture arguments. Ivanov [16] develops the odd-characteristic signed mechanism and its two-colour version; Mghirbi–Douglas [23] establish the qualitative result used above and the deterministic cone criterion. We prove the needed probability and image estimates for our different finite populations, and include the adapted bounded-path and cone arguments in the appendices.

Bunina, Lee, Miasnikov, and Muliarchyk [6] obtain one-sided inverses in group algebras in every odd prime characteristic using projective-plane line types, additional labels, and direction-dependent multiplicities with integer corrections to balance inverse-label slots. Their finite construction differs from the Hadamard auxiliary blocks, two-pencil residue pairing, and coordinatewise line–union repair used here. Their root-noncollapse argument does not require torsion-freeness, whereas the torsion-free hyperbolic realization in this paper uses the cone criterion of Mghirbi–Douglas.

The compact exceptional root and an exact integer-lattice balancing procedure are already present in Mghirbi–Douglas [23, Section 7.2]. The new finite design instead combines Hadamard auxiliary blocks, a residue pairing over $\mathbb{Z}/p$ obtained from two pencils, coordinatewise line–union repair, and union types outside the replicated bulk. Classical ingredients include Singer's indexing of projective lines [32] and Paley's constructions of Hadamard matrices [30]; their use here yields the quantitative bounds, not new Hadamard-existence results. The source of the general existence bounds is discussed in Section 5.

In characteristic three, the cyclic plane of order 81 and eleven auxiliary blocks with Hadamard parameters of order 12 give the explicit incidence data; the two-pencil seed and fixed repair-only union rows supply the exact balances. Table 2 compares the two direct-finiteness constructions, including their degree trade-off; the 6648-letter zero-divisor-only model in [23, Section 6] is not the appropriate comparison. The support restrictions of Appendix D concern collected group-algebra coefficients rather than incidence alphabets. None of these alphabet estimates bounds the final replication required by the realization theorem.

1.4. **Conventions.** A signed alphabet T has a fixed-point-free involution ι. Write $\iota z(t) = z(\iota(t))$ for incidence vectors. A T-labelled graph is immersed when its outgoing labels at each vertex are distinct. Opposite orientations of an edge carry inverse letters. Girth counts loops and parallel pairs as cycles of lengths one and two. Length counts traversals, not distinct image edges. All graph roots are prescribed before the matching is drawn. Indexed copies of a type are distinct vertices; no injectivity of the final vertex-to-group-label map is presumed. Entrywise squaring is denoted $\circ 2$. All logarithms are natural except $\log_2$. Every asymptotic estimate keeps the entire finite input fixed, including the characteristic, and varies only replication.

2. Incidence data and the finite interface

The finite input must do two jobs before any edge is chosen. Its row intersections encode the signed algebraic identities, while its integer populations make the required inverse-letter matchings possible. Conditions (F1) and (F2) express these requirements. Conditions (F3) and (F4) control labelled paths: the first bounds candidate intermediate vertices at every finite

size, and the second gives decay of squared word weights. The probabilistic and geometric consequences will be derived from these conditions in Sections 7–8.

Definition 2.1 (Admissible finite matching input)**.** Fix a prime p. The input consists of a signed alphabet (T, ι), normal subsets $S_i \subseteq T$, an exceptional subset S_r, and affine nonnegative integer multiplicities for three classes A_m^+, A_m^-, B_m, with exactly one exceptional vertex r_A in A_m^+. A normal vertex $r_B \in B_m$ is also prescribed. There are fixed integers $N_0, F_0 > 0$, $m_0 \geq 1$, constants C_{aff}, D, and a kernel $\Theta : T^2 \to [0, 1)$ such that, for $m \geq m_0$ and $n = mN_0$, the following hold.

(F1) All normal-row inner products are 1 modulo p, including self-products. The exceptional row has inner product 1 with every normal row and self-product 0 modulo p.

(F2) Every listed multiplicity is positive. Each of the three classes has size $n + O(C_{\mathrm{aff}})$. Put $f = F_0/N_0$. Every one-letter pool has size $nf + O(C_{\mathrm{aff}})$, with absolute error at most C_{aff}, and

$$|A_{m,t}^+| = |A_{m,\iota(t)}^-|, \qquad |B_{m,t}| = |B_{m,\iota(t)}|.$$

Every vertex has between 82 and D outgoing slots.

(F3) For each reduced turn $u \neq \iota(t)$ and each class $Y \in \{A_m^+, A_m^-, B_m\}$,

$$|\{x \in Y : \iota(t), u \in S_x\}| \leq nf\Theta(t, u).$$

Here $\Theta(t, \iota(t)) = 0$, every reduced value is positive, and $\Theta(t, u) = \Theta(\iota(u), \iota(t))$.

(F4) There are $1 \leq \varphi(t) < \infty$ and $0 < \Lambda < 1$ such that $\Theta^{\circ 2}\varphi \leq \Lambda\varphi$.

The constants, types, affine offsets, kernel and roots are fixed before m varies. The $O(C_{\mathrm{aff}})$ class-size bounds in (F2) also mean absolute error at most C_{aff}.

The graph is formed only after these data are fixed: for each inverse pair choose two independent bijections $A_{m,t}^+ \to A_{m,\iota(t)}^-$ and $A_{m,t}^- \to A_{m,\iota(t)}^+$, and one bijection $B_{m,t} \to B_{m,\iota(t)}$. A matched pair of slots is an edge with its two inverse orientations. Different bijections have disjoint slot sets, though their endpoint vertices need not be disjoint. This produces immersed graphs Γ_A, Γ_B, and Γ_A is bipartite.

2.1. **Ordinary projective rows.** Lines in a projective plane give the intersection congruences with no choice of coefficients: unequal lines meet once, while the size of a line is also one modulo the characteristic. We also need rows that permit local integer adjustments without disturbing these congruences. Unions of concurrent lines provide precisely such rows. Let $q = p^r$ and $v = q^2 + q + 1$. Identify the points $P = \mathrm{PG}(2, q)$ with $\mathbb{Z}/v$ using a Singer action [32]. One construction is multiplication on the one-dimensional subspaces of $\mathbb{F}_{q^3}$ by a generator of $\mathbb{F}_{q^3}^\times/\mathbb{F}_q^\times$. This action is regular on points and on lines: a nonidentity multiplier has degree three over $\mathbb{F}_q$, so has no nonzero proper invariant subspace. Write L_x for the translates of one line. Then

$$|L_x| = q + 1, \quad |L_x \cap L_y| = 1 \ (x \neq y), \quad \sum_x \mathbf{1}_{L_x} = (q+1)\mathbf{1}_P. \tag{2.1}$$

Choose $p + 1$ distinct lines L_{ℓ_j} through 0 and put $R_x = x + \bigcup_j L_{\ell_j}$. Outside their common point these lines are disjoint. Thus, over the integers,

$$p\mathbf{e}_x = \sum_{j=1}^{p+1} \mathbf{1}_{L_{x+\ell_j}} - \mathbf{1}_{R_x}, \qquad |R_x| = k_R := (p+1)q + 1. \tag{2.2}$$

In particular each union row is a sum of $p + 1 \equiv 1 \pmod p$ line rows modulo p. Equation (2.1) and bilinearity show that every line/union inner product is 1 modulo p.

The cyclic arithmetic is only an incidence index. It does not impose the relation $t^v = 1$ on any generator of the eventual group.

2.2. **Hadamard auxiliary rows.** An auxiliary block must meet every other auxiliary block in a multiple of the characteristic, including itself. For the matching estimates we also want each auxiliary letter, and each pair of distinct auxiliary letters, to occur equally often. A normalized Hadamard matrix gives both kinds of regularity: its two Gram identities control block intersections and point replication separately.

Lemma 2.2 (Normalized Hadamard core)**.** *Let H be a normalized real Hadamard matrix of order m_H, with first row and column equal to* 1*. Put*

$$H = \begin{pmatrix} 1 & \mathbf{1}^{\mathsf{T}} \\ \mathbf{1} & C_H \end{pmatrix}, \qquad e = m_H - 1, \quad h = m_H/2, \quad B_H = (J - C_H)/2.$$

The rows of B_H define e blocks H_j on e auxiliary letters E. Every block has size h, distinct blocks intersect in $h/2$ letters, every letter lies in h blocks, and every distinct pair of letters lies in $h/2$ blocks.

Proof. The Hadamard identities on both sides imply

$$C_H\mathbf{1} = C_H^{\mathsf{T}}\mathbf{1} = -\mathbf{1}, \qquad C_H C_H^{\mathsf{T}} = C_H^{\mathsf{T}} C_H = m_H I - J.$$

Consequently B_H is a zero-one matrix with row and column sums h, and

$$B_H B_H^{\mathsf{T}} = B_H^{\mathsf{T}} B_H = \frac{m_H}{4}(I + J). \tag{2.3}$$

The two products give, respectively, block intersections and point-pair replication. □

Assume $4p \mid m_H$. Then e is odd, $e \equiv -1 \pmod p$, and $h, h/2$ are divisible by p. No cyclic structure on E is required. In particular the residue calculations in the next section take place in $\mathbb{Z}/p$, not in $\mathbb{Z}/e$.

Use the following normal types, with $x \in P$ and $1 \le j \le e$:

$$L_x, \qquad L_x \cup H_j, \qquad R_x. \tag{2.4}$$

Only undecorated unions are used; their role is integer repair rather than the leading distribution. The exceptional type is

$$S_r = L_0 \cup E. \tag{2.5}$$

Lemma 2.3 (Incidence congruences)**.** *For the types* (2.4)–(2.5)*, every normal-row inner product is* 1 *modulo p. The exceptional row has inner product* 1 *with every normal row and satisfies $|S_r| \equiv 0 \pmod p$. Their degrees lie between $q+1$ and*

$$D = \max\{(p+1)q + 1,\ q + 1 + e\}. \tag{2.6}$$

Proof. Equation (2.2) proves the ordinary congruences. Auxiliary products contribute $0, h$, or $h/2$, all zero modulo p. Intersecting the root with a normal row contributes the line intersection and either 0 or h auxiliary letters. Finally $q + 1 + e \equiv 0 \pmod p$. A decorated line has degree $q + 1 + h \le q + 1 + e$. □

3. Two pencils and exact integer repair

Adding the exceptional root changes the slot counts at the coordinates of S_r. To restore the inverse-letter balances, we seek integer row combinations X, Y with $X + \iota Y = \mathbf{1}_{S_r}$. We choose ι by first solving a residue-pairing problem, and then use the coordinate identity (2.2) to remove the remaining defect over the integers. The two-pencil calculation depends on the number of auxiliary letters, not on an auxiliary-prime realization of the blocks.

To see the required residue counts, begin with a line combination w supported on P. Its auxiliary coordinates are zero, so each of the e auxiliary letters must be paired with an ordinary coordinate of residue one. After these pairs are removed, the ordinary residues must pair under $t \mapsto 1 - t$. Writing n_t for their counts, we therefore require $n_1 - n_0 = e$, equality of the other complementary counts, and even cardinality in the fixed residue class 2^{-1}.

Two affine pencils turn the pairing problem into a short calculation with residue histograms. Each affine point belongs to exactly one line of each pencil, so its coefficient is the sum of

two pencil coefficients and its histogram is their cyclic convolution. Points at infinity behave differently: each pencil centre sees the sum of all coefficients in its pencil, while the remaining points see only the coefficient on the common line at infinity. Thus the affine convolution alone cannot determine the final residue counts.

For $p \geq 5$ we choose the first histogram to differ from uniformity at just two residues, 0 and $k = 4^{-1}$. Convolution with this difference turns the required affine correction into the cyclic difference equation $\beta_t - \beta_{t-k} = C_t$. To integrate it into an *integer* histogram, the total increment must vanish; to choose the solution with sum zero, its first moment must also vanish modulo p. The second moment then governs the coefficient sum at the pencil centres. These conditions leave room to adjust the two complementary residue classes adjacent to the centres. Concentrating the correction C at four residues gives the explicit solution below: its 1-versus-0 imbalance supplies the e auxiliary partners, and the remaining entries compensate for the two centres and the points at infinity. This explains the form of the coefficients without requiring any special arithmetic structure on the auxiliary letters.

3.1. **A residue seed for $p \geq 5$.** Represent residues by integers in $\{0, \dots, p-1\}$. Put

$$h_0 = 2^{-1}, \qquad k = 4^{-1}, \qquad c = 1 - k \quad \text{in } \mathbb{Z}/p.$$

The five residues $0, k, h_0, c, 1$ are distinct. Define integer vectors α, C on $\mathbb{Z}/p$ by

$$\alpha_0 = 1, \quad \alpha_k = -1, \qquad C_1 = e, \quad C_k = 2, \quad C_c = 4, \quad C_{h_0} = -e - 6, \tag{3.1}$$

with all other entries zero.

The relations $h_0 = 2k$, $c = 3k$ and $1 = 4k$ make the moment calculation particularly short. Lemma 3.1 solves the difference equation; Lemma 3.2 then shows how the two pencil centres complete the required complementary counts.

Lemma 3.1 (Cyclic coefficient construction)**.** *There is a unique integer vector β with*

$$\beta_t - \beta_{t-k} = C_t, \qquad \sum_t \beta_t = 0. \tag{3.2}$$

It satisfies

$$\alpha * \beta = C, \qquad \sum_t t\beta_t \equiv -k \pmod{p}, \qquad \max_t |\beta_t| \leq e + 6.$$

Proof. Using $h_0 = 2k$, $c = 3k$, $1 = 4k$ and $e \equiv -1 \pmod{p}$ gives

$$\sum_t C_t = 0, \qquad \sum_t tC_t \equiv 0, \qquad \sum_t t^2 C_t \equiv 2k^2 \pmod{p}.$$

Integrate the difference equation around the k-cycle. The zero total makes the resulting integer solution periodic. Its first moment gives $0 \equiv -k \sum_t \beta_t$, so its sum is divisible by p. Subtracting its integer mean centers it, uniquely. The second moment gives

$$2k^2 \equiv -2k \sum_t t\beta_t - k^2 \sum_t \beta_t,$$

which proves the moment assertion. Convolution with α is exactly the difference equation. The total positive variation of the cyclic increments C_t is $e + 6$. Hence the range of an integrated solution has length at most $e + 6$; its mean lies between its minimum and maximum, proving the last bound after centering. □

Choose $q = p^r \geq 81$ with

$$q/p + \alpha_t \geq 0, \qquad q/p + \beta_t \geq 0 \quad (t \in \mathbb{Z}/p). \tag{3.3}$$

A sufficient condition is $q/p \geq e + 6$. Put $F_t = q/p + \alpha_t$ and $G_t = q/p + \beta_t$. Both histograms sum to q. Choose maps $f_0, g_0 : \mathbb{F}_q \to \{0, \dots, p-1\}$ with these histograms, using any fixed enumeration. In homogeneous coordinates set

$$w = \sum_{x \in \mathbb{F}_q} f_0(x) \mathbf{1}_{\{X = xZ\}} + \sum_{y \in \mathbb{F}_q} g_0(y) \mathbf{1}_{\{Y = yZ\}} + h_0 \mathbf{1}_{\{Z=0\}}, \qquad w|_E = 0. \tag{3.4}$$

All coefficients here are nonnegative integers; h_0 is its chosen representative. These are actual projective lines, each of which has a unique Singer index.

Lemma 3.2 (Complementary residues)**.** *The ordinary residue counts of w are*

$$n_t = q^2/p + C_t + 2[t = k] + (q-1)[t = h_0]. \tag{3.5}$$

They satisfy $n_1 - n_0 = e$, $n_k = n_c$, equality of every other pair n_t, n_{1-t} with $t \neq 0, 1$, and evenness of n_{h_0}. There is therefore a fixed-point-free involution ι on $T = P \sqcup E$, with $\iota(E) \subset P$, such that

$$w_t + w_{\iota(t)} \equiv 1 \pmod{p}. \tag{3.6}$$

Proof. At an affine point $(x, y, 1)$ the value is $f_0(x) + g_0(y)$, with histogram $F * G = q^2/p + C$. The sum of the coefficients in either affine pencil is $-k$ modulo p. The uniform contribution is $q(p-1)/2$, which is divisible by p. Thus each pencil center has residue $-k + h_0 = k$, and each other point at infinity has residue h_0. This proves (3.5).

The stated complementary counts follow from (3.1). Since q^2/p and $e + 6$ are odd, $n_{h_0} = q^2/p - e - 6 + q - 1$ is even. Pair the e auxiliary letters with e ordinary residue-one coordinates; pair the remaining complementary residue classes, and pair the h_0 class internally. The first step is possible because $n_1 = n_0 + e$. Ordering each class makes the pairing deterministic. □

3.2. **Lifting the defect over the integers.** The residue pairing makes each remaining coordinate defect divisible by p. The line–union identity can therefore correct it exactly. We first obtain the full-alphabet defect, then replace the full root by the compact root using an integer expression for the all-ordinary row. Let $\mathcal{L} \subseteq \mathbb{Z}^T$ be the integer span of the undecorated line and union rows. For $z \in \mathbb{Z}^T$ the expression ιz denotes coordinate permutation, not a permutation of row coefficients.

Proposition 3.3 (Coordinate repair and compact root)**.** *Suppose $w \in \mathcal{L}$ is supported on P and satisfies* (3.6)*. There are explicitly determined integer combinations $X, Y \in \mathcal{L}$ with*

$$X + \iota Y = \mathbf{1}_{S_r}, \qquad S_r = L_0 \cup E. \tag{3.7}$$

Proof. Set $z_t = (1 - w_t - w_{\iota(t)})/p \in \mathbb{Z}$. Begin with $X_T = Y = w$. For every ordinary t, add $pz_t\mathbf{e}_t$ to X_T using (2.2). For every auxiliary a, add $pz_a\mathbf{e}_{\iota(a)}$ to Y using the same identity. No auxiliary coordinates are added to either row expansion. If $t, \iota(t) \in P$, only the X_T correction at t affects $X_T(t) + Y(\iota(t))$. If $\iota(t) \in E$, that sum is $w_t + pz_t = 1$; and on an auxiliary a it is $w_{\iota(a)} + pz_a = 1$. Hence $X_T + \iota Y = \mathbf{1}_T$.

Summing (2.2) and using (2.1) yields

$$\mathbf{1}_P = \left(1 - \frac{(p+1)q}{p}\right) \sum_x \mathbf{1}_{L_x} + \frac{q}{p} \sum_x \mathbf{1}_{R_x}. \tag{3.8}$$

The coefficients are integers. Take $X = X_T - \mathbf{1}_P + \mathbf{1}_{L_0}$ to obtain (3.7). □

Write $X = \sum_i x_i \mathbf{1}_{S_i}$ and $Y = \sum_i y_i \mathbf{1}_{S_i}$ using (2.4), with zero coefficients on decorated rows. These expansions are algorithmic: each coordinate correction changes $p + 1$ line coefficients and one union coefficient. Consequently their sizes can be bounded before any reservoir or bulk replication is chosen.

For the nonnegative seed (3.4), let

$$C_{\text{base}} = \sum_t t(F_t + G_t) + h_0, \qquad Z_* = \frac{2(q+1)C_{\text{base}} - (v+e)}{p}. \tag{3.9}$$

Every $w_t + w_{\iota(t)}$ is a nonnegative integer congruent to 1, so it is at least 1. Thus $z_t \leq 0$ and $\sum_t |z_t| = Z_*$. The full-root repair has coefficient norm, including one for the root, at most

$$1 + 2C_{\text{base}} + (p+2)Z_*.$$

Passing to the compact root adds at most

$$v\left(\left|1 - \frac{(p+1)q}{p}\right| + \frac{q}{p}\right) + 1$$

to this bound. The smaller root thus trades vertex degree against the size of the integer correction; it need not decrease the repair norm. For subsequent estimates define the actual finite quantities

$$B_* := \max\{1, \max_i |x_i|, \max_i |y_i|\}, \qquad C_{\mathrm{rep}} := 1 + \sum_i |x_i| + \sum_i |y_i|. \tag{3.10}$$

They do not depend on replication. The five distinct residues used above are unavailable at $p = 3$. Section 6 uses a simpler seed in that case, followed by exactly Proposition 3.3.

4. A fixed reservoir and uniform turn domination

The repair identity fixes the required corrections but not the vertex populations. We place lines and Hadamard-decorated lines in the replicated bulk and keep unions in a balanced finite reservoir. In this way the arithmetic corrections remain available at every finite size without contributing to the leading turn distribution. The remaining candidate-count error is bounded independently of replication, so a strict spectral margin in the leading kernel can absorb it. Theorem 4.1 expresses that margin through two inequalities in an equitable quotient of the full squared-turn kernel.

Theorem 4.1 (Hadamard input with fixed incidence repair)**.** *Let $p \geq 5$ be prime, let a real Hadamard matrix of order m_H divisible by $4p$ be given, and put*

$$e = m_H - 1, \quad h = m_H/2, \quad q = p^r \geq 81, \quad v = q^2 + q + 1, \quad f = (q+1)/v.$$

Assume (3.3)*. Suppose a rational $r_0 \geq 1$ satisfies*

$$R_1 = \frac{(e-1)/4 + [v + e(r_0 - 1)]f^2}{r_0} < 1, \tag{4.1}$$

$$R_3 = \frac{v - 1 + e(r_0 - 1)}{(q+1)^2} + ef^2 < 1. \tag{4.2}$$

Then there is admissible finite matching input with $v + e$ signed letters, root $L_0 \cup E$, degree bound (2.6)*, and primitive leading bulk and slot counts*

$$N_0 = (h/g)v^2, \qquad F_0 = (h/g)v(q+1), \qquad g = \gcd(h, q+1). \tag{4.3}$$

Only lines and decorated lines occur in the leading bulk. All union multiplicities are fixed. Positivity and finite-size turn domination hold above explicit finite thresholds determined below. The same conclusion holds at $p = 3, q = 81, e = 11$ with the seed of Section 6.

First note that the hypotheses imply $e < v$. Indeed (4.2) and $r_0 \geq 1$ imply

$$\frac{q}{q+1} + e\frac{(q+1)^2}{v^2} < 1, \quad \text{so} \quad e < \frac{v^2}{(q+1)^3} < v.$$

Thus the three classes used below are well defined and the root degree bound is meaningful. Existence of the involution also follows directly from Lemma 3.2.

4.1. **The primitive leading distribution.** Choose the ordinary line uniformly. Independently of that choice, assign probability f/h to each particular auxiliary block and the remaining probability to the empty auxiliary part. Since an auxiliary letter lies in h blocks, its frequency is f, the same as that of an ordinary letter. Clearing denominators and dividing by the common divisor gives the following integer table. For each line L_x, use

$$b_0 = \frac{hv - e(q+1)}{g} \quad \text{copies of } L_x, \qquad b_1 = \frac{q+1}{g} \quad \text{copies of each } L_x \cup H_j. \tag{4.4}$$

There are no leading union copies. These are integers. Since $e/h < 2$ and $v > 2(q+1)$, $b_0 > 0$. Their sum is N_0. An ordinary point is on $q+1$ line classes, each containing hv/g vertices. An auxiliary point is on h blocks in each of v line classes. Both have F_0 slots, proving (4.3).

This table is primitive: $\gcd(v, q+1) = 1$ gives

$$\gcd(hv - e(q+1), q+1) = \gcd(h, q+1) = g,$$

so $\gcd(b_0, b_1) = 1$. Primitivity is only for this uniform product-distribution table, not a minimum over all finite designs. The auxiliary choice is independent of the ordinary line; its empty-part probability is $1 - ef/h > 0$. At most one auxiliary block is chosen.

4.2. **An exactly balanced reservoir.** The union coefficients in the repair can have either sign, so union types must be available on both sides even though their leading multiplicities vanish. We add a finite population with the same slot count at every letter. It supplies enough union copies for positivity without introducing a new inverse-letter imbalance. The line copies allow the ordinary slot count to be matched by auxiliary decorations. Let the repair coefficients be those of Proposition 3.3, and put $A_* = hv - e(q+1) > 0$. Choose

$$r_R = h(B_* + 1), \qquad \ell_R = h\left(1 + \left\lceil \frac{er_R k_R}{hA_*} \right\rceil\right). \tag{4.5}$$

Take r_R undecorated union copies and ℓ_R line copies at every translate. The ordinary slot count is

$$F_R = \ell_R(q+1) + r_R k_R, \qquad N_R = v(\ell_R + r_R). \tag{4.6}$$

It is divisible by h. Decorate F_R/h line copies by each auxiliary block. The required eF_R/h decorations fit among the $v\ell_R$ line copies because

$$eF_R/h \le v\ell_R \quad \Longleftrightarrow \quad er_R k_R \le \ell_R A_*.$$

The ceiling in (4.5) proves this. Assign consecutive decoration requests cyclically to the v line indices, and at each index use the next unused line copy. No index receives more than $\lceil eF_R/(hv) \rceil \le \ell_R$ requests. Every auxiliary coordinate then receives $h(F_R/h) = F_R$ slots, exactly as every ordinary coordinate does. Denote the resulting nonnegative row multiplicities by r_i.

The reservoir multiplicities and both repair arrays are now independent of replication. The repairs are offsets to the complete populations, not necessarily removals from reservoir copies alone: an undecorated line may draw on its leading copies. For $n = mN_0$ prescribe

$$\begin{array}{l|l} A_m^+ & mb_i + r_i - x_i \text{ normal copies, and one root of type } S_r \\ A_m^- & mb_i + r_i + y_i \text{ normal copies} \\ B_m & mb_i + r_i \text{ normal copies.} \end{array} \tag{4.7}$$

On unions $b_i = 0$, but $r_R > B_*$ makes their multiplicities strictly positive. On undecorated lines positivity holds when $mb_0 > B_*$. Decorated-line repairs vanish and $b_1 > 0$. Thus

$$m_{\mathrm{pos}} = 1 + \lfloor B_*/b_0 \rfloor \tag{4.8}$$

is sufficient. A smaller directly checked reservoir or positivity threshold may be used.

The slot vectors are

$$\begin{aligned} \mu_+ &= (mF_0 + F_R)\mathbf{1} - X + \mathbf{1}_{S_r}, \\ \mu_- &= (mF_0 + F_R)\mathbf{1} + Y, \qquad \mu_B = (mF_0 + F_R)\mathbf{1}. \end{aligned} \tag{4.9}$$

Equation (3.7) proves the exact inverse-letter balances. The class sizes are

$$|A_m^+| = n + N_R - \sum_i x_i + 1, \quad |A_m^-| = n + N_R + \sum_i y_i, \quad |B_m| = n + N_R. \tag{4.10}$$

A safe common bound for class offsets and pool offsets is

$$C_{\mathrm{aff}} := N_R + C_{\mathrm{rep}}. \tag{4.11}$$

For pools this uses $F_R \le N_R$ and $|X(t)| \le \sum |x_i|$, $|Y(t)| \le \sum |y_i|$. In particular $|V\Gamma| \le 3n + 3C_{\mathrm{aff}} < 4n$ when $n > 3C_{\mathrm{aff}}$. Fix r_B as the first undecorated L_0 vertex, before choosing any edges.

4.3. **The leading reduced-turn kernel.** A path arriving with letter t and leaving with letter u requires the two outgoing coordinates $\iota(t), u$ at its intermediate vertex. The leading weight is their joint frequency divided by the common one-letter frequency f. This normalization is what remains after the intermediate-vertex count is combined with the matching-pool denominator. For $u \neq \iota(t)$ define

$$w_0(t,u) = \begin{cases} (q+1)^{-1}, & \iota(t), u \in P, \\ f, & \text{exactly one of } \iota(t), u \text{ belongs to } E, \\ 1/2, & \iota(t), u \in E, \end{cases} \qquad w_0(t, \iota(t)) = 0. \tag{4.12}$$

The number of leading vertices containing the two outgoing coordinates $\iota(t), u$ is exactly $nfw_0(t,u)$. Two ordinary points lie on one line, giving joint frequency $1/v$ and conditional weight $1/(q+1)$. Ordinary and auxiliary inclusion are independent, giving joint frequency f^2 and conditional weight f. Two auxiliary points lie together in $h/2$ blocks, each selected with probability f/h, whereas one point lies in h blocks. Their joint frequency is therefore $f/2$ and their conditional weight is $1/2$. In particular the auxiliary entry of the squared-turn kernel is $1/4$, with no additional factor of f^2.

Let

$$B_{\mathrm{coord}} = \max_{t \in T} \max \left\{ \sum_i |x_i|[t \in S_i],\ \sum_i |y_i|[t \in S_i] \right\}, \qquad E_* = F_R + B_{\mathrm{coord}} + 1. \tag{4.13}$$

For any two distinct coordinates, their reservoir incidence is at most F_R; their absolute correction incidence is at most B_{coord}; the root costs at most one. Therefore, uniformly in all reduced turns and in all three classes,

$$|\{x : \iota(t), u \in S_x\}| \leq nfw_0(t,u) + E_*. \tag{4.14}$$

The constants E_* and C_{aff} have different roles and need not be comparable.

The reduced-turn weight at (t,u) depends on the *incoming coordinate* $\iota(t)$ and the outgoing coordinate u. Since each auxiliary letter is paired by ι with an ordinary letter, the partition relevant to rows of the kernel is not simply $P \sqcup E$. It is

$$\iota(E), \qquad E, \qquad P \setminus \iota(E).$$

For $t \in \iota(E)$ the incoming coordinate is auxiliary, so its row has the auxiliary–auxiliary weights; for $t \in E$ it is one of the paired ordinary coordinates; and for the remaining t it is an unpaired ordinary coordinate. Backtracking removes a different target in each case. These three classes therefore give an exact equitable quotient, and a positive class-constant vector can certify contraction on the entire signed alphabet.

Lemma 4.2 (Full-alphabet contraction)**.** *The equitable quotient of $w_0^{\circ 2}$ on $\iota(E), E, P \setminus \iota(E)$ is*

$$Q = \begin{pmatrix} ef^2 & (e-1)/4 & (v-e)f^2 \\ (e-1)/(q+1)^2 & ef^2 & (v-e)/(q+1)^2 \\ e/(q+1)^2 & ef^2 & (v-e-1)/(q+1)^2 \end{pmatrix}. \tag{4.15}$$

On the vector $(r_0, 1, 1)$ the first and third ratios are (4.1)–(4.2); the second is $R_3 - (r_0-1)/(q+1)^2$. If $R = \max\{R_1, R_3\} < 1$, then the lifted vector

$$\varphi(t) = r_0 \quad (t \in \iota(E)), \qquad \varphi(t) = 1 \quad \textit{otherwise}$$

satisfies $w_0^{\circ 2}\varphi \leq R\varphi$.

Proof. For $t \in \iota(E)$ the incoming coordinate $\iota(t)$ is auxiliary. The v ordinary successors have weight f, and the $e-1$ nonbacktracking auxiliary successors have weight $1/2$. For $t \in E$, the incoming coordinate lies in $\iota(E)$, so exactly one target in that class is excluded. For $t \in P \setminus \iota(E)$ the excluded target lies in that same class. These observations give the three rows. The class-constant subspace is invariant, so its inequality lifts coordinatewise to the entire alphabet. □

The quotient bounds concern the ideal leading population. The actual population contributes at most the fixed additive error in (4.14); increasing replication makes this error small relative to every positive leading candidate count. We reserve part of the strict contraction margin by setting

$$\tau = (1-R)/8, \qquad \Lambda = (1+R)/2, \qquad \Theta = (1+\tau)w_0. \tag{4.16}$$

Writing $u = 1-R \in (0,1)$ shows $(1+u/8)^2(1-u) < 1-u/2 = \Lambda$. Thus $\Theta^{\circ 2}\varphi < \Lambda\varphi$. Reduced values of w_0 lie between $1/(q+1)$ and $1/2$, and $\tau < 1/8$, so $0 < \Theta < 1$ on reduced turns. Reversal symmetry follows from symmetry of the pair-incidence counts. Finally

$$m \geq m_{\mathrm{turn}} := \left\lceil \frac{E_*(q+1)}{F_0\tau} \right\rceil \quad \Longrightarrow \quad nfw_0(t,u) + E_* \leq nf\Theta(t,u). \tag{4.17}$$

Thus one kernel bounds all reduced-turn candidate counts in every class, uniformly above a single finite threshold. Pointwise convergence of frequencies alone would not provide this bound. Equations (4.8) and (4.17), together with Lemma 2.3, prove Theorem 4.1.

Remark 4.3 (Why the fixed types are retained). The union vertices are present in every finite graph, and their contributions are included in (4.14). Their leading frequencies may vanish because every letter pool remains linear in n and every reduced leading turn is strictly positive. A new letter with zero leading pool, or a new positive turn at a zero entry of the reduced leading kernel, would not be covered by this argument. Positivity of every type's limiting frequency is not used in the realization estimates of Section 7.

5. Plane order p^3 and uniform alphabet bounds

For a given Hadamard order, the size of the projective plane is constrained in two independent ways: the residue histograms must be nonnegative, and the squared-turn kernel must contract. Theorem 4.1 makes both constraints explicit. A general sufficient inequality handles large auxiliary orders; at order $4p$ a sharper test vector already permits $q = p^3$.

Changing the auxiliary architecture matters here. The observation after Mghirbi–Douglas's Lemma 7.8 identifies a failure of their displayed sufficient inequality at $q = p^4$ [23]. Proposition C.1 goes further for that unchanged model: its squared-turn kernel itself fails to contract. The smaller planes below therefore use a different finite input, not merely a sharper test vector for their kernel.

Lemma 5.1. *Let $e \geq 19$, $q \geq 81$, and $q \geq 8(e^2+e)$. For $r_0 = e+1$, the ratios in* (4.1)–(4.2) *satisfy*

$$R_1 < \frac{1}{3}, \qquad R_3 \leq 1 - \frac{3}{4q} < 1.$$

Proof. Since $f = (q+1)/(q^2+q+1) < 1/q$ and $vf^2 = 1 + q/v < 1 + 1/q$,

$$R_1 < \frac{(e+3)/4 + 1/q + e^2/q^2}{e+1} \leq \frac{(e+3)/4 + 9/(8q)}{e+1} < \frac{1}{3}.$$

For the last inequality it suffices that $27/(8q) < (e-5)/4$, which follows from $q \geq 81$ and $e \geq 19$. Also $1/(q+1) \geq 7/(8q)$, and therefore

$$R_3 = 1 - \frac{1}{q+1} + \frac{e^2}{(q+1)^2} + ef^2 \leq 1 - \frac{7}{8q} + \frac{e^2+e}{q^2} \leq 1 - \frac{3}{4q}.$$

□

5.1. Hadamard matrices of order $4p$.

At this order the auxiliary alphabet grows linearly with p. A test vector adapted to that smaller alphabet gives the following direct estimate.

Theorem 5.2. *Let $p \geq 5$ be prime. If a real Hadamard matrix of order $4p$ exists, then Theorem 4.1 applies with*

$$e = 4p-1, \qquad q = p^3, \qquad r_0 = p + \frac{3}{4}.$$

The signed alphabet has size p^6+p^3+4p, the root has degree p^3+4p, the maximum matching-graph degree is $(p+1)p^3+1$, and the primitive leading bulk is

$$N_0 = p(p^6+p^3+1)^2.$$

Proof. Lemma 3.1 gives $\|\beta\|_\infty \le e+6 = 4p+5 \le p^2 = q/p$, so both histograms are nonnegative. We verify the sharper spectral conditions directly, rather than imposing the sufficient condition of Lemma 5.1. Here

$$e(r_0-1) = (4p-1)(p-1/4) < 4p^2.$$

Using $vf^2 < 1+1/q$ gives

$$r_0R_1 < p+\frac{1}{2}+\frac{1}{p^3}+\frac{4}{p^4} < p+\frac{3}{4} = r_0.$$

For $p \ge 5$, the last two fractions are at most $1/125+4/625 < 1/4$. Similarly,

$$R_3 < 1-\frac{1}{p^3+1}+\frac{4p^2+4p}{p^6} < 1.$$

Indeed the last inequality is equivalent to $4(p+1)(p^3+1) < p^5$, and

$$\frac{4(p+1)(p^3+1)}{p^5} = \frac{4(p+1)}{p^2}\left(1+\frac{1}{p^3}\right) \le \frac{24}{25}\frac{126}{125} = \frac{3024}{3125} < 1.$$

Thus Theorem 4.1 applies. Since $h = 2p$ and $\gcd(2p, p^3+1) = 2$, its primitive bulk is pv^2. Finally $e = 4p-1 < pq$, so the union degree, not the root degree, is maximal. □

Table 1 records instances obtained from actual Paley matrices. The Paley type II instances have composite auxiliary alphabet sizes; no prime of size e is required. The exact matrix construction is specified in Appendix C. The accompanying program checks every matrix entry, both Hadamard-design Gram identities, the coefficient histograms, and the rational Perron inequalities. These finite checks are separate from the projective-plane existence theorem and the later selection of a matching.

TABLE 1. Order-$4p$ inputs with $q = p^3$. The last column counts signed letters.

p	m_H	e	q	Hadamard construction	$\lvert T\rvert$
5	20	19	125	Paley I over $\mathbb{F}_{19}$	15 770
7	28	27	343	Paley II over $\mathbb{F}_{13}$	118 020
11	44	43	1 331	Paley I over $\mathbb{F}_{43}$	1 772 936
13	52	51	2 197	Paley II over $\mathbb{F}_{25}$	4 829 058
17	68	67	4 913	Paley I over $\mathbb{F}_{67}$	24 142 550
19	76	75	6 859	Paley II over $\mathbb{F}_{37}$	47 052 816

5.2. **The Ghaderpour parameter bound.** For a uniform statement, an existence theorem must replace the assumed order-$4p$ matrix. We first retain the parameter choice arising from Ghaderpour's bound, and then use Du–Jiang's smaller orders directly. The two statements below share the same geometric construction. Ghaderpour's Theorem 10 states the following one-variable specialization [14, pp. 14–15 of the preprint]. The logarithms in that section are to base two. For a positive integer $u > 1$, there is an integer $N(u)$ with

$$N(u) \le \frac{1}{5}\log_2(u-1)+14$$

such that a full real orthogonal design of order and type $2^a u$ exists for every integer $a \ge N(u)$. Setting its sole variable to one gives a real Hadamard matrix of that order. Thus the endpoint of the cited theorem is a real Hadamard matrix, not merely a matrix over a signed group. Ghaderpour's proof uses the complementary-sequence estimates of Livinskyi [19, Chapter 5], following the signed-group method of Craigen [8]. The exact orders used below also follow directly from Du–Jiang's theorem and Sylvester doubling, as we show in the proof. This alternative

avoids reliance on the older sequence estimate; Du–Jiang discuss a difficulty with one of its antecedents in [10, p. 2, footnote 1].

Theorem 5.3. *For every prime $p \geq 2^{60}$, there is admissible finite input with $q = p^3$ and*

$$|T| = p^6 + p^3 + p2^{a_p} \leq p^6 + p^3 + 2^{15}p^{6/5}, \qquad a_p = 14 + \left\lceil \tfrac{1}{5}\log_2(p-1) \right\rceil. \tag{5.1}$$

Its maximum matching-graph degree is $(p+1)p^3 + 1$.

Proof. First, the order $m_H = p2^{a_p}$ can be obtained without using the complementary-sequence estimate in Ghaderpour's proof. Put $t_p = 6\lfloor \log_2 p/40 \rfloor + 10$. The Du–Jiang theorem (5.2) supplies a Hadamard matrix of order $p2^{t_p}$. Since $p - 1 \geq p/2$,

$$a_p \geq 14 + \tfrac{1}{5}\log_2(p-1) \geq \tfrac{69}{5} + \tfrac{1}{5}\log_2 p > 10 + \tfrac{3}{20}\log_2 p \geq t_p.$$

Tensoring with a Sylvester matrix of order $2^{a_p - t_p}$ gives the exact required order. In particular it is divisible by $4p$, and

$$m_H \leq 2^{15}p(p-1)^{1/5} < 2^{15}p^{6/5}.$$

Put $e = m_H - 1$ and $q = p^3$. Then

$$8e(e+1) < 8m_H^2 < 2^{33}p^{12/5} \leq p^3,$$

because $p^{3/5} \geq 2^{36}$. Furthermore $e + 6 = m_H + 5 \leq 2^{16}p^{6/5} \leq p^2$, since $p^{4/5} \geq 2^{48}$. Hence the histograms are nonnegative. Lemma 5.1 and Theorem 4.1 apply with $r_0 = e + 1$. Now $v + e = p^6 + p^3 + m_H$, giving (5.1). Since $e < p^2$, the union degree dominates the root degree. □

5.3. **A refinement using Du–Jiang.** The preceding proof used Sylvester doubling to reach a prescribed order. Using the Du–Jiang order without that extra doubling improves both the lower-order alphabet term and the range of characteristics. Du and Jiang prove that, for every positive odd integer u, there is a real block-circulant Hadamard matrix of order

$$u2^{6\lfloor \log_2 u/40 \rfloor + 10}. \tag{5.2}$$

This is [10, Theorem 3, p. 18]; the proof, including its finite complementary-sequence coverage input, is in their Section IV, with Table II on p. 26. We use this existence theorem, not its matrix-generation code, and no block-circulant property is required of our auxiliary design.

Theorem 5.4. *For every prime $p \geq 2^{23}$, there is admissible finite input with $q = p^3$ and*

$$|T| = p^6 + p^3 + p2^{6b_p+10} \leq p^6 + p^3 + 2^{10}p^{23/20}, \qquad b_p = \lfloor \log_2 p/40 \rfloor. \tag{5.3}$$

Its maximum matching-graph degree is $(p+1)p^3 + 1$.

Proof. Set $b = b_p$, $m_H = p2^{6b+10}$, and $e = m_H - 1$. The order in (5.2) is divisible by $4p$. We claim

$$p \geq 2^{12b+23}, \qquad p \geq 2^{6b+11}.$$

If $b = 0$, both follow from $p \geq 2^{23}$. If $b \geq 1$, then $p \geq 2^{40b}$; the inequalities follow from $40b \geq 12b + 23$ and $40b \geq 6b + 11$. Consequently

$$8e(e+1) < 8m_H^2 = p^2 2^{12b+23} \leq p^3, \qquad e + 6 = m_H + 5 \leq p^2/2 + 5 \leq p^2.$$

Histogram feasibility and Lemma 5.1 again apply. Theorem 4.1 gives the input at $q = p^3$. Finally $2^{6b} \leq p^{3/20}$ proves (5.3), and $e < p^2$ gives the degree assertion. □

Theorem 5.3 retains the parameter choice suggested by Ghaderpour's stated bound, with an alternative proof of the needed orders from Du–Jiang. Theorem 5.4 uses the latter orders directly and improves both the explicit range and the lower-order alphabet term. Neither supplies a replication parameter uniform in the characteristic.

Corollary 5.5. *Admissible finite input exists for every odd prime.*

Proof. For $p \geq 5$, use a Hadamard matrix supplied by (5.2) and choose a sufficiently large power q of p so that $q \geq 81$, $q \geq 8(e^2 + e)$, and $q/p \geq e + 6$. Lemma 5.1 and Theorem 4.1 apply. Characteristic three is treated next. □

One can alternatively choose a prime $e \equiv -1 \pmod{4p}$ by Dirichlet's theorem [9] and use a Paley matrix of order $e+1$. This recovers the auxiliary-prime route to the finite input, but with the fixed reservoir of Section 4. Its relevance is an elementary explicit auxiliary design, not a better uniform bound for the least permissible prime. The matrix-existence routes above remove that arithmetic restriction.

6. A complete characteristic-three input

Characteristic three provides a small concrete instance of the whole construction. Its eleven auxiliary blocks have sizes and pair intersections that can be checked directly, and only three residue classes need to be paired. We use the cyclic plane of order 81, a two-pencil residue seed, and fixed-multiplicity union rows outside the replicated bulk. The resulting alphabet has 6654 signed letters. Table 2 compares this input with Mghirbi–Douglas's displayed direct-finiteness construction.

6.1. A two-pencil residue seed.

Here the residue classes zero and one must differ in size by eleven, while the class two must have even size. These conditions admit a direct two-pencil seed, without the five-residue construction used for larger characteristics.

Lemma 6.1. *Let $q = 3^r \geq 9$. In $\mathrm{PG}(2,q)$ choose distinct points o_0, o_1 on a line L_∞. On the other q lines through o_0 and through o_1, assign coefficients $0,1,2$ with respective histograms*

$$F = (q/3+1, q/3-1, q/3), \qquad G = (q/3-3, q/3+1, q/3+2).$$

Give L_∞ coefficient 2, and let w be the resulting integer sum of line indicators. The ordinary residue counts are

$$(n_0, n_1, n_2) = (q^2/3 - 5,\ q^2/3 + 6,\ q^2/3 + q). \tag{6.1}$$

In particular $n_1 - n_0 = 11$, and n_2 is even.

Proof. The deviations from the uniform histograms are $(1,-1,0)$ and $(-3,1,2)$. Their cyclic convolution modulo three is $(-5,4,1)$. Each affine point is the intersection of exactly one line of each pencil; its value is the sum of those two coefficients. Its residue histogram is therefore $(q^2/3-5, q^2/3+4, q^2/3+1)$. At each center the coefficients sum to 1 modulo three: the two pencil sums are $q-1$ and $q+5$, respectively, and the common line adds 2. The other $q-1$ points of L_∞ have value 2. Adding these contributions gives (6.1). The last entry is even because $q^2/3$ and q are odd. □

For the concrete instance put $q = 81$, $v = 6643$, $P = \mathbb{Z}/v$, and let $\mathcal{D}$ be the ordered 82-element difference set in Appendix C. Its ordered nonzero differences are precisely the nonzero elements of P. Thus $L_x = \mathcal{D} + x$ gives all lines. The chosen pencils are through the points 0 and 1, and their common line is L_0. Order the line indices in each pencil increasingly, omitting 0, and assign consecutive coefficient classes with histograms

$$F = (28, 26, 27), \qquad G = (24, 28, 29).$$

Add coefficient 2 on L_0. This determines 111 nonzero line coefficients with sum 168. Lemma 6.1 gives residue counts $(2182, 2193, 2268)$.

Take $E = \mathbb{Z}/11$ disjoint from P and the blocks

$$H_j = E \setminus (\{1,3,4,5,9\} + j), \qquad j \in \mathbb{Z}/11.$$

They have size 6; distinct blocks intersect in 3 points; each point lies in 6 blocks and each distinct pair in 3. These are the Hadamard parameters of order 12, also verified directly from the eleven sets. Set $w|_E = 0$. Pair the ordered extra letters with the first eleven residue-one points of P; pair the remaining residue-one and residue-zero points in increasing order; and pair consecutive residue-two points. This is a specified fixed-point-free involution ι with $w + \iota w \equiv 1 \pmod 3$.

Use the four concurrent-line shifts

$$(\ell_1, \ell_2, \ell_3, \ell_4) = (0, 6642, 6640, 6634)$$

to define R_x. The pencil identity is $\sum_{j=1}^{4} \mathbf{1}_{L_{x+\ell_j}} - \mathbf{1}_{R_x} = 3\mathbf{e}_x$, and $|R_x| = 325$. Form X_T, Y as in Proposition 3.3. In this instance

$$\mathbf{1}_P = -107 \sum_x \mathbf{1}_{L_x} + 27 \sum_x \mathbf{1}_{R_x}, \qquad X = X_T - \mathbf{1}_P + \mathbf{1}_{L_0}. \tag{6.2}$$

Then $X + \iota Y = \mathbf{1}_{L_0 \cup E}$. All these instructions use integer arrays of length 6643 or 6654 and fixed finite sums; a finite-field implementation or a random seed is unnecessary.

6.2. **Bulk, reservoir, and distinct error bounds.** The leading table is the primitive product distribution from Section 4. For this explicit repair a smaller reservoir than the general sufficient choice can be checked directly. We record separately the error in a two-coordinate candidate count and the error in the class and pool sizes, since the matching estimates use both. The leading table has, at every line index, 19478 undecorated copies and 41 copies with each H_j. Thus

$$N_0 = 132388347, \qquad F_0 = 1634178, \qquad f = 82/6643. \tag{6.3}$$

For the reservoir take two line copies and four undecorated union copies at every index. For each $0 \le j < 11$, decorate one copy of every line with index $244j + k$, $0 \le k < 244$, by H_j. These 2684 indices are distinct and less than 6643, so at least one line copy at every index remains undecorated. The reservoir has

$$N_R = 39858, \qquad F_R = 2 \cdot 82 + 4 \cdot 325 = 1464 = 244 \cdot 6 \tag{6.4}$$

slots at every letter. This is a smaller explicitly verified reservoir than the general sufficient choice (4.5).

The coefficient data reconstructed from the two pencils give

$$\begin{aligned} \sum_i x_i &= 510897, & \sum_i y_i &= -18, \\ C_{\mathrm{rep}} &= 1 + \sum_i |x_i| + \sum_i |y_i| = 856270, \\ B_{\mathrm{coord}} &= 16952, & E_* &= F_R + B_{\mathrm{coord}} + 1 = 18417, \\ C_{\mathrm{aff}} &= N_R + C_{\mathrm{rep}} = 896128. \end{aligned} \tag{6.5}$$

The coefficient ranges on undecorated lines and unions are

$$52 \le x_L \le 107, \quad -27 \le x_R \le 2, \qquad -54 \le y_L \le 2, \quad 0 \le y_R \le 29.$$

Consequently all multiplicities in (4.7) are positive for $m \ge 1$. In particular, each positive union multiplicity is at least $4 - 2 = 2$, and each negative union multiplicity is at least 4.

The class cardinalities, with $n = mN_0$, are exactly

$$|A_m^+| = n - 471038, \qquad |A_m^-| = n + 39840, \qquad |B_m| = n + 39858. \tag{6.6}$$

Every pool is exactly balanced with its required inverse pool by (4.9). The turn-count error E_* bounds the candidate count for a specified pair of outgoing coordinates. It is not a bound for the affine class offsets: $471038 > 18417$. The separate constant C_{aff} controls classes and pools in the switching estimates. Since $N_0 > 3C_{\mathrm{aff}}$, the convenient bound $|V\Gamma| < 4n$ holds for all $m \ge 1$.

These finite numerical statements can be verified without enumerating pairs of graph vertices. Expand the coefficient arrays over all translates of $\mathcal{D}$ and R_0, compute their sums and absolute sums, and check $X(t) + Y(\iota(t)) = \mathbf{1}_{L_0 \cup E}(t)$ at each letter. Apply the same expansion to the absolute coefficient arrays to obtain B_{coord}. The specification in this section and Appendix C is complete; the supplied certificate stores the derived arrays for a second comparison.

6.3. **An exact finite-size spectral certificate.** The remaining task is to certify the turn bound at an actual finite replication, rather than only in the leading distribution. The rational row ratios below give a contraction margin; the coordinate error then determines the stated sufficient threshold.

Proposition 6.2. *The data above satisfy Definition 2.1 for all $m \geq 925$, with root degree 93, minimum degree 82, maximum degree $D = 325$, and*

$$\Theta = (1 + 1/1000)w_0, \qquad \varphi|_{\iota(E),E,P\setminus\iota(E)} = (15/4, 1, 1), \qquad \Lambda = 249/250.$$

Here w_0 is the exact leading kernel (4.12).

Proof. The incidence and matching assertions follow from the preceding construction and Proposition 3.3. The only normal degrees are $82, 88, 325$; the unique root has degree $82+11 = 93$. Lemma 4.2 gives the exact raw row ratios

$$\frac{206926074}{220647245}, \qquad \frac{589637388083}{593452830152}, \qquad \frac{1179760200105}{1186905660304}.$$

The last is the largest. Multiplication by $(1001/1000)^2$ gives a maximum of

$$\frac{28550196842541}{28665756800000} = \frac{249}{250} - \frac{896930259}{28665756800000} < \frac{249}{250}.$$

This proves the full-alphabet inequality $\Theta^{\circ 2}\varphi < \Lambda\varphi$. Since the minimum reduced weight is $1/82$, finite-size domination holds when

$$\frac{mF_0}{82000} \geq E_* = 18417.$$

At $m = 925$ the difference between the left side and E_* is $693/40 > 0$; at $m = 924$ it is $-651/250 < 0$. Hence 925 is the least integer certified by this particular sufficient bound. Reversal symmetry and $0 < \Theta < 1$ are immediate from (4.12). □

TABLE 2. Displayed characteristic-three direct-finiteness inputs. The comparison concerns these specified constructions, not optimality.

Quantity	Mghirbi–Douglas [23]	Present model
Signed letters	59306	6654
Exceptional-root degree	257	93
Maximum graph degree	257	325

Mghirbi–Douglas use $q = 3^5 = 243$ and $e = 3^2 + 3 + 1 = 13$. Their normal degrees are 244 and 253, and their root has degree 257 [23, Definition 7.1, Lemma 7.2, and Section 7.4]. The present construction has a smaller alphabet and root degree, but its maximum graph degree 325 exceeds their 257; the comparison is not componentwise.

Remark 6.3 (Interpretation of the finite bounds). The alphabet has 6654 signed letters, and the rose has 3327 generators. Neither number is a coefficient support size or a relator count. The root degree is 93 and the maximum matching-graph degree is 325. The primitive leading bulk N_0 is a normalization before replication; the reservoir adds 39858 vertices to each leading class before the signed repairs.

The compact-root choice has a cost within the two-pencil construction: its repair norm 856270 exceeds the corresponding full-root norm 34807. This comparison concerns the specified incidence choices, not optimality among all designs.

The threshold $m \geq 925$ concerns turn domination only. With the girth scale of Section 7, $b_{\mathrm{scale}} = 18$ and $A = 4608$, so even $L \geq 3$ requires $n \geq 2^{13824}$. No terminal group-defining matching is claimed at $m = 925$, and the finite input bounds do not estimate computational running time.

7. Conditioned matchings and the realization estimates

Fix finite input from Theorem 4.1 or Proposition 6.2. All types, affine corrections and roots are now deterministic; randomness enters only through the compatible slot bijections as replication grows. The realization requires matchings of large girth, and it requires probability estimates *under that conditioning*. A switching argument supplies uniform bounds on prescribed edges within the actual sign subpools. From those bounds we obtain expansion and, more importantly for the bounded-path theorem, counts of labelled injective images.

The switching and expansion arguments follow [28, Section 3.3, Lemma 3.3 and Proposition 3.4], with the prescribed colour data of [16, Construction 4.1 and Section 6.1]; compare also [23, Proposition 7.9]. Their classical switching antecedents are [21, 22]. The task here is to establish the required counts for the affine populations, including the fixed-multiplicity union types.

Write $\mathbb{P}$ for the independent uniform law on the compatible bijections of Definition 2.1. We consider only m beyond the finite positivity and turn-domination thresholds, and subsequently beyond $n > 3C_{\mathrm{aff}}$. Set

$$b_{\mathrm{scale}} = \min\{b \in \mathbb{N} : 2^b \geq 8D/f\}, \qquad A = 256b_{\mathrm{scale}}, \qquad L = \left\lfloor \frac{\log_2 n}{A} \right\rfloor, \qquad n = mN_0. \tag{7.1}$$

Let $\mathcal{G}_L$ denote the event that $\Gamma = \Gamma_A \sqcup \Gamma_B$ has girth at least L.

7.1. **Why maximum degree, rather than alphabet size, suffices.** The finite type of every prospective vertex is prescribed before edges are sampled. At that vertex there are at most D outgoing slots. Thus a description of an oriented cycle can choose its consecutive vertices first, and then choose at most D outgoing labels at each chosen vertex. It need not choose from every letter of T at every step. Similarly, among a prescribed set of $2k$ vertices, an oriented matched pair is specified by its source vertex, one of at most D source slots, and its target vertex, giving at most $4Dk^2$ possibilities. An invalid target slot only removes a possibility.

These are counting statements about prescribed slots, not about word entropy. Word counts in Appendix A still use the full alphabet. This distinction justifies using $D = 325$, rather than $|T| = 6654$, in (7.1) and in the expansion estimates for characteristic three.

Lemma 7.1 (A switch within one subpool)**.** *Let xy and uz be oriented edges belonging to the same matching bijection. Suppose uz is on no cycle shorter than L, and the distance between $\{x, y\}$ and $\{u, z\}$ exceeds $2L$. Replacing xy, uz by xz, uy preserves all slots and creates no cycle shorter than L.*

Proof. The operation transposes two targets of one bijection, preserving types and, on A, the two sign classes. A new short cycle containing only one new edge would give an old path of length less than L between the two distant endpoint sets. If it contains both new edges, remove them. The remaining paths either join the distant sets, or join x to y and u to z. The latter u–z path, together with uz, is an old cycle shorter than L. Both possibilities are excluded. The argument allows $x = y$, in which case the corresponding residual path may be empty. □

Proposition 7.2. *For all sufficiently large m, $\mathcal{G}_L$ is nonempty.*

Proof. A feasible collection of k distinct prescriptions has unconditioned probability given by reciprocals of falling factorials in the actual subpools, and hence at most $(nf - C_{\mathrm{aff}} - k)^{-k}$. A cycle of length k has at most two colourings on A, and one side choice on B. By the slot count above its number of vertex/label descriptions is at most $3(n + C_{\mathrm{aff}})^k D^k$. A simple cycle prescribes k distinct edges, including loops and parallel pairs at lengths one and two. Thus the expected number of edges lying on cycles shorter than L is at most

$$3 \sum_{k<L} k(4D/f)^k = o(n).$$

Indeed $4D/f < 2^{b_{\mathrm{scale}}}$ and (7.1) make the exponential factor at most $n^{1/256}$.

Choose an outcome with $o(n)$ such bad edges. There are $O(D^{2L+2}) = o(n)$ edges incident to the radius-$2L$ neighborhoods of the endpoints of any specified edge; here $D^{2L} \leq n^{1/128}$. Every subpool has $nf + O(1)$ edges. Therefore any bad edge has, in its own subpool, a distant edge

which is not bad. Lemma 7.1 removes that bad edge without creating another, and any old short cycle not using it is unchanged. Iteration ends after finitely many steps at an outcome in $\mathcal{G}_L$. □

Define $\mathbb{P}_L = \mathbb{P}(\cdot \mid \mathcal{G}_L)$ for these m. A prescription refers to an undirected matching edge; writing it in reverse does not make a new prescription.

Lemma 7.3 (Conditioned prescriptions)**.** *There is $r_n = C_{\mathrm{aff}} + O(D^{2L+2}) = o(n)$ such that, for every set S of s distinct prescriptions with $\mathbb{P}_L(S) > 0$ and every further prescription $e \notin S$,*

$$\mathbb{P}_L(e \mid S) \le \frac{1}{nf - s - r_n} \tag{7.2}$$

when the denominator is positive. In particular, uniformly over sets of $E \le C_1 L$ distinct prescriptions, for fixed C_1,

$$\mathbb{P}_L(\textit{all prescriptions}) \le (nf)^{-E} \exp(o(L)). \tag{7.3}$$

Proof. The conditional law is uniform on a finite set of tuples satisfying S. From a tuple containing $e = xy$, at least $nf - s - r_n$ edges of its subpool are unprescribed and far from x, y. All edges are now on no cycle shorter than L. Switch with any such uz. The switch preserves S, deletes e, and stays in $\mathcal{G}_L$ by Lemma 7.1. The output determines the input and the switch uniquely: in its bijection ψ, one recovers $z = \psi(x)$ and $u = \psi^{-1}(y)$. Counting these injective images gives (7.2). A zero-probability exposure prefix gives zero joint probability; otherwise successive application of the bound gives

$$(nf)^{-E} \exp\left(O\left(\frac{E(E + r_n)}{n}\right)\right),$$

which is (7.3) for $E = O(L)$. No estimate from below for $\mathbb{P}(\mathcal{G}_L)$, or independence after conditioning, is used. □

7.2. **Expansion and component diameter.** A small set with too few neighbours would force many prescribed edges into a small vertex set. The conditioned prescription bound makes such configurations unlikely. Once these sets expand, disjoint-ball packing along a geodesic gives a logarithmic diameter bound. Set

$$\rho = \left(\frac{f}{1000D}\right)^2, \qquad d_0 = \lceil 1000A/\rho \rceil, \qquad D_0 = d_0 + 32A. \tag{7.4}$$

Proposition 7.4. *With $\mathbb{P}_L$-probability tending to one, every component of Γ has diameter at most $d_0 L$.*

Proof. Suppose a vertex set S of size $k \le \rho n$ has a closed neighborhood of size less than $2k$. Some set Z of $2k$ vertices contains that neighborhood. The minimum degree is at least 82, so Z contains at least $41k$ distinct edges incident to S. Since the whole graph has fewer than $4n$ vertices, there are at most $(2\mathrm{e}n/k)^{2k}$ choices for Z. There are at most $4Dk^2$ candidate prescriptions on it. Take $r = 41k$. At each positive-probability prefix of an r-edge prescription, (7.2) has denominator at least $nf/2$ for all large n, since $41\rho < f/4$ and $r_n = o(n)$. Using $\mathrm{e} < 3$, the union bound is at most

$$\left(C_3 (k/n)^{39}\right)^k, \qquad C_3 = 36\left(\frac{24D}{41f}\right)^{41}. \tag{7.5}$$

The chosen ρ satisfies

$$C_3 \rho^{39} = 36(24/41)^{41} 1000^{-78} (f/D)^{37} < \tfrac{1}{2}.$$

For $k \le \sqrt{n}$ sum the upper bounds $(C_3 n^{-39/2})^k$; for larger k sum 2^{-k}. Both sums tend to zero. Thus every set of size at most ρn doubles on adjoining its neighbors, with probability tending to one.

On this event a ball of radius $R = \lceil \log_2 n \rceil + 2$ has more than ρn vertices, since it doubles until passing that threshold. Along a geodesic, radius-R balls centered $2R + 1$ apart are disjoint. There are fewer than $4/\rho$ such balls because $|V\Gamma| < 4n$. The resulting diameter bound is less than $(4/\rho)(2R + 1)$. For all large n, (7.1) bounds this by $d_0 L$. □

7.3. **Word decay and the model-dependent image estimate.** The finite kernel now enters in two ways. Its entrywise-square contraction bounds the number of heavily weighted words, and its candidate-count interpretation bounds injective realizations of a fixed labelled image. These are the two estimates used at every multiplicity level in Appendix A. For a word $w = t_1 \cdots t_s$, define

$$P_\Theta(w) = \prod_{j<s} \Theta(t_j, t_{j+1}).$$

It is zero on words with immediate backtracking. The kernel symmetry $\Theta(t,u) = \Theta(\iota(u), \iota(t))$ gives the combined inversion-and-reversal identity

$$P_\Theta(t_1 \cdots t_s) = P_\Theta(\iota(t_s) \cdots \iota(t_1)),$$

that is, $P_\Theta(w^{-1}) = P_\Theta(w)$. Fix

$$\gamma = (1-\Lambda)/8, \qquad a_0 = q+1, \qquad \varepsilon = \frac{\gamma}{32(1+a_0)}. \tag{7.6}$$

For all our inputs, $a_0 > -\log \min_{u \neq \iota(t)} \Theta(t,u)$, since the minimum is at least $1/(q+1)$. Any larger a_0 could instead be used, with the corresponding smaller ε.

Lemma 7.5. *For all sufficiently large s, depending only on the fixed finite input,*

$$\sum_{w \in T^s} P_\Theta(w)^2 \leq \exp(-2\gamma s).$$

Proof. For $M = \Theta^{\circ 2}$ the sum is $\mathbf{1}^{\mathsf{T}} M^{s-1} \mathbf{1} \leq (\sum_t \varphi(t)) \Lambda^{s-1}$. Since $\log \Lambda \leq -(1-\Lambda)$ and $2\gamma = (1-\Lambda)/4$, the fixed prefactor is absorbed for sufficiently large s. □

The passage from finite incidence data to the inherited bounded-path argument occurs in Lemma 7.6. In a labelled chain, an internal vertex is chosen from a two-coordinate candidate pool, while each distinct edge consumes one matching prescription. The two counts are normalized by the same *letter pool*, so their powers cancel. The sides and, in A, the bipartition signs must be kept fixed for this cancellation to be uniform. Unlike an estimate normalized by a vertex-type frequency, the result continues to hold when a union type occurs only in the fixed reservoir; compare [28, equation (4.11)] and [23, proof of Proposition 7.9(c)].

Lemma 7.6 (Injective image estimate)**.** *Let Δ be an abstract finite graph without isolated vertices, decomposed into Q chains with disjoint degree-two interiors and marked endpoints. Let it have $E \leq C_1 L$ edges and V vertices, with fixed immersed edge labels. Prescribe its component sides, and proper signs on its A-part. If $k_0 \in \{0,1\}$ marked vertices are prescribed to their corresponding fixed root, then*

$$\mathbb{E}_L \#\{\textit{injective realizations of } \Delta\} \leq n^{V-E-k_0} f^{-Q} \prod_{\textit{internal chain turns}} \Theta(t,u)\, \exp(o(L)). \tag{7.7}$$

The error is uniform for fixed C_1, Q.

Proof. There are $E-Q$ internal chain vertices. Definition 2.1(F3) bounds their candidate counts by $nf\Theta(t,u)$ on their prescribed sign class or B. Each other unfixed marked vertex has at most $n + C_{\mathrm{aff}}$ choices, and a prescribed root has one. The product of these candidate counts is at most

$$n^{V-k_0} f^{E-Q} \prod \Theta(t,u) \exp(O(L/n)).$$

Retain only injective assignments compatible with all slots. Each requires E distinct prescriptions, so Lemma 7.3 gives (7.7). Assignments identifying vertices do not represent this image graph and are not counted as independent edge prescriptions. A connected bipartite image has only two sign colourings; recording them costs a bounded factor per component, not one factor per edge. □

All reservoir vertices have already been included in the finite candidate bounds. A fixed root contributes exactly one possible image, and every denominator is a positive letter pool rather than a limiting vertex-type frequency. These are the features needed to apply the multiplicity argument of Appendix A to the present distribution.

7.4. **Bounded-path exclusion.** Cone pictures will compare long portions of paths whose labels agree, possibly after reversing orientation. These paths may overlap or revisit edges, so a bound based only on independently exposed traversals would not apply. The relevant objects record occurrences and comparisons, but impose distinctness on the underlying edges in each compared pair. A path system is a family of positive-length graph paths, all closed and cyclically immersed except possibly one immersed interval starting at r_A or r_B. A comparison matches two equal-length intervals of occurrences, either in order with equal letters or in reverse order with inverse letters. All compared occurrences are disjoint, and each matched pair must traverse distinct *undirected* graph edges. Intervals in closed paths may be cyclic. Let H be total traversal length and b_0 the number of unpaired occurrences.

Theorem 7.7 (Bounded-path estimate for the fixed-reservoir model)**.** *With ε fixed by* (7.6)*, for every fixed integer $K \geq 1$, real $C \geq 1$, and integer $I \geq 0$, the $\mathbb{P}_L$-probability that there is a path system with at most K paths, $L \leq H \leq CL$, at most I comparisons, and $b_0 \leq \varepsilon H$, tends to zero as $m \to \infty$.*

The proof is given in Appendix A. It is the multiplicity-stage argument of [28, Theorem 4.1, Lemmas 4.2–4.4], with Ivanov's prescribed colour data [16, Section 6.1] and the image estimate (7.7). Repeated edge traversals are counted at multiplicity levels, never as independent matching prescriptions. The common approximate grid, projections of complete compatible assignments and minimum of stage first-moment bounds are inherited parts of that argument. Their details are included to make both the reduction to the present finite model and the order $\exists\varepsilon\,\forall K, C, I$ explicit.

8. Selection, hyperbolicity, and group-algebra witnesses

The construction now has the incidence congruences for the algebraic identities and the path estimates needed for realization. Three geometric conclusions remain. Asphericity gives a finite-dimensional classifying complex and hence torsion-freeness; root protection makes the second witness nonzero; a linear area bound gives hyperbolicity. Mghirbi–Douglas's deterministic criterion supplies all three from a pattern-free graph [23, Theorem B].

The distinction between its three picture problems matters. Spherical pictures test asphericity, rooted pictures test paths with prescribed endpoints in the graph, and free-boundary pictures test arbitrary null words. The last case is needed for the area bound: exclusion of reduced spheres alone would not establish it. We state the parameters before choosing the one finite test used to select a matching.

8.1. **The deterministic criterion.** For $\varepsilon \in (0, 1]$ and $d, \lambda_{\mathrm{geo}} \geq 1$, set

$$
\begin{aligned}
C_{\mathrm{sep}} &= \frac{2\sqrt{2}}{1-\sqrt{2/3}}, & D_{MD} &= \lceil d + 4\lambda_{\mathrm{geo}} + 8 \rceil\,, \\
U_{MD} &= \max\{3, \lceil 96 D_{MD}/\varepsilon \rceil\}, & \eta_{MD} &= \min\{1/(32U_{MD}), \varepsilon/(64U_{MD})\}, \\
K_{MD} &= \max\{1, \left\lceil (C_{\mathrm{sep}}/\eta_{MD})^2 \right\rceil\}, & C_{MD} &= K_{MD}(U_{MD} + D_{MD}), \quad I_{MD} = \lceil 192 C_{MD} \rceil\,.
\end{aligned} \tag{8.1}
$$

A finite rooted immersed labelled graph is *pattern-free* at scale L when: its minimum degree is at least five and its girth is at least $L \geq 3$; all component diameters are at most dL and $\log_2 |V\Gamma| \leq \lambda_{\mathrm{geo}} L$; and it has no path system of the kind in Theorem 7.7 for the triple (K_{MD}, C_{MD}, I_{MD}). Both I_{MD} and our I count comparison *pairs*, not individual interval sides.

Theorem 8.1 (Mghirbi–Douglas's cone criterion)**.** *For a pattern-free graph Γ, attach an abstract cone on every component to the rose by the label map, and call the resulting complex X. Then X is aspherical. No path in Γ from a designated root to a different vertex has trivial label in $G = \pi_1(X)$. With respect to the spanning-tree presentation, every word trivial in G satisfies*

$$
\mathrm{Area}(w) \leq \frac{16}{13\varepsilon}|w|. \tag{8.2}
$$

In particular G is finitely presented, torsion-free and word-hyperbolic.

This is [23, Definition 3.3 and Theorem B, pp. 9–16]. Appendix B records its proof and the relation to the OpenAI cone-picture and planar-extraction arguments. Linear isoperimetry implies hyperbolicity by [4, Theorem III.H.2.9]. No assumption that an attached cone embeds in X is made.

8.2. **A single finite selection test.** Bounded-path exclusion fixes ε before the size parameters of a path system. We can therefore choose one triple large enough for the deterministic cone criterion and then let replication grow. No simultaneous probabilistic estimate over all triples is required. Using ε from (7.6) and D_0 from (7.4), choose, in this order,

$$U = \max\{3, \lceil 96D_0/\varepsilon \rceil\}, \quad \eta = \frac{\varepsilon}{64U}, \quad K = \left\lceil (20/\eta)^2 \right\rceil, \quad C = K(U+D_0), \quad I = \lceil 192C \rceil . \tag{8.3}$$

These are fixed before replication. Define $\mathcal{E}_m$ to be the set of matching tuples having girth at least L, all component diameters at most d_0L, and no forbidden path system for (8.3).

Proposition 8.2 (Hyperbolicity transfer)**.** *For every sufficiently large m, $\mathcal{E}_m$ is nonempty. If $\Gamma \in \mathcal{E}_m$, $L \geq 3$, and $|V\Gamma| < 4n$, then Γ is pattern-free for*

$$d = d_0, \qquad \lambda_{\mathrm{geo}} = 2A,$$

and Theorem 8.1 applies with the same ε.

Proof. Proposition 7.2 first defines the girth-conditioned law. Proposition 7.4 and Theorem 7.7, applied to the one fixed triple (8.3), show that $\mathbb{P}_L(\mathcal{E}_m) \to 1$. Hence it is positive for all sufficiently large m.

For any tuple as stated, minimum degree and girth are sufficient for the first pattern-free condition. The vertex-count condition follows from

$$\log_2 |V\Gamma| < 2 + \log_2 n < 2 + A(L+1) \leq 2AL,$$

where the last inequality holds for $L \geq 3$ and $A \geq 1$. The selected diameter is d_0L. Finally

$$D_{MD} = d_0 + 8A + 8 \leq d_0 + 32A = D_0, \qquad C_{\mathrm{sep}} = 6\sqrt{2} + 4\sqrt{3} < 20.$$

Since $\varepsilon \leq 1$, the minimum defining η_{MD} is $\varepsilon/(64U_{MD})$. Monotonicity gives successively

$$U_{MD} \leq U, \quad \eta_{MD} \geq \eta, \quad K_{MD} \leq K, \quad C_{MD} \leq C, \quad I_{MD} \leq I.$$

Thus every MD-forbidden system is among those excluded by our selection. Its possible rooted interval and its distinct-undirected-edge restriction are precisely the conventions of Theorem 7.7; allowing cyclic intervals in our closed paths only enlarges the tested family. All three conditions of the criterion are verified. □

Remark 8.3 (An explicit area constant). For Proposition 6.2, (7.6) gives $\gamma = 1/2000$, $a_0 = 82$, and $\varepsilon = 1/5312000$. Thus

$$\mathrm{Area}(w) \leq \frac{84992000}{13}|w| < 6537847|w|.$$

The more conservative choice $a_0 = 100$ instead gives $\varepsilon = 1/6464000$ and the area bound $\mathrm{Area}(w) \leq (103424000/13)|w| < 7955693|w|$. The parameter comparison in Proposition 8.2 works for either choice. These are bounds for a selected presentation, not estimates of a least Dehn constant over all presentations of the group.

The selection can be made algorithmic. Fix enumerations for fields, lines, type copies and roots, and lexicographic choices for every finite construction. List m increasingly, and discard values which fail positivity, finite-size turn domination, $n > 3C_{\mathrm{aff}}$ or $L \geq 3$. Enumerate the finitely many compatible matching permutations at each remaining m, and test girth, component diameters and the fixed BP triple. The BP test lists finitely many graph paths of total length at most CL, their interval comparisons, and their labels; it does not solve a word problem in an as-yet unconstructed group. Proposition 8.2 proves that some tuple passes. Choose the first. This is a terminating specification, with no practical complexity bound.

8.3. **The algebraic identities.** For a selected matching, the cone criterion supplies the only geometric facts needed in the cancellation calculation: closed component paths have trivial label, and the designated roots are protected. We sum over complete root components so that the simultaneous-edge operation stays within the summation sets. For a selected tuple let

$$X = F \cup_\Gamma \coprod_{\Lambda \in \pi_0(\Gamma)} \mathrm{Cone}(\Lambda), \qquad G = \pi_1(X), \tag{8.4}$$

where F is the rose on T/ι. Let A' and B' be the complete components of r_A, r_B. The label of a root-to-vertex path defines $g_x \in G$ for $x \in A'$ and $h_y \in G$ for $y \in B'$. These do not depend on the path because the cones kill the labels of every closed path. For an outgoing edge labelled t,

$$g_{x\cdot t} = g_x t, \qquad h_{y\cdot t} = h_y t, \qquad g_{r_A} = h_{r_B} = 1.$$

Let $\sigma(x) = 1$ on A_m^+ and -1 on A_m^-. Every A-edge reverses this sign.

The sign rule has an earlier counterpart in Mineyev's signature condition [24, Section 2.2]; the distinction, explained in [23, Section 10, "Signed schemes"], is what is paired. A product structure pairs product terms themselves. Here we pair a multiset of darts, in which the term $g_x h_y^{-1}$ appears $|S_x \cap S_y|$ times. The integer identity cancels that multiset, and the incidence congruences convert it into the required product identity over the prime field.

Lemma 8.4 (Signed dart identity)**.** *For complete components U, V, possibly equal, with path labels g_x, h_y and an alternating sign σ on U, one has in $\mathbb{Z}[G]$*

$$\sum_{x \in U,\, y \in V} |S_x \cap S_y| \sigma(x) g_x h_y^{-1} = 0. \tag{8.5}$$

Proof. On the finite set of triples (x, y, t) with $t \in S_x \cap S_y$, the map

$$(x, y, t) \longmapsto (x \cdot t, y \cdot t, \iota(t))$$

is a fixed-point-free involution. It preserves $g_x h_y^{-1}$ and reverses $\sigma(x)$. Summing the paired terms proves (8.5). This is the signed simultaneous-step mechanism of [16, Lemma 2.2], used here also with $U = V$ as in [23, Section 7]; the exceptional-root assembly comes from [28, Proposition 2.1]. □

Theorem 8.5 (Realization of the finite constructions)**.** *Each input constructed in Theorem 4.1 or Proposition 6.2 has a selected realization for which G is finitely presented, torsion-free, and word-hyperbolic, with a finite classifying complex of dimension at most two. Define*

$$a = \sum_{x \in A'} \sigma(x) g_x, \qquad b = \sum_{x \in A'} g_x^{-1}, \qquad c = \sum_{y \in B'} h_y^{-1} \quad \text{in } \mathbb{F}_p[G]. \tag{8.6}$$

Then $ab = 1$, $ac = 0$, $c \neq 0$, and $ba \neq 1$. The group admits a presentation on $|T|/2$ alphabet generators satisfying (8.2)*.*

Proof. The probabilistic estimates and Proposition 8.2 give a tuple to which Theorem 8.1 applies. Thus (8.4) is a finite aspherical complex, the two roots are protected, and the group has all the asserted geometric properties.

Apply (8.5) to $A' \times A'$. The incidence congruences are

$$|S_x \cap S_z| \equiv 1 - [x = z = r_A] \pmod p.$$

The root has positive sign and label one. Reducing (8.5) modulo p therefore gives $0 = ab - 1$. No vertex of B' is exceptional, so all intersections on $A' \times B'$ are one modulo p; the same identity gives $ac = 0$. By root protection, $h_y = 1$ only for $y = r_B$. Hence the coefficient of the identity in c is exactly one, and $c \neq 0$. If $ba = 1$, then $c = b(ac) = 0$, a contradiction.

For the finite presentation choose a spanning tree in each component of Γ. Use as relators the labels of its fundamental cycles associated to nontree edges. The cone is homotopy equivalent relative to its attaching graph to killing that free basis by discs, so this gives the group of (8.4). The generating alphabet is T/ι, with $|T|/2$ generators. Its area convention is exactly that of Theorem 8.1. □

The algebraic identity does not require different vertices or different simultaneous-step components to represent distinct group elements. Coincident values are simply collected after their signed contributions have been summed. This is also why the support of a witness cannot be read from the number of graph vertices.

A one-sided inverse with a nonzero annihilated element gives the usual ring-theoretic consequences below. The implication from a torsion-free direct-finiteness counterexample to the other Kaplansky counterexamples is noted in [3, Section 1]; the explicit square-zero and unit calculations are also recorded in [23, Lemmas 2.1–2.2].

Corollary 8.6. *For each realization above and every field K of characteristic p, the algebra $K[G]$ is neither directly finite nor stably finite. It has a nontrivial idempotent, a nonzero square-zero element, and a unit of order p not of the form αg, with $\alpha \in K^\times$ and $g \in G$. Moreover G is not sofic.*

Proof. Put $e_0 = ba$. Then $e_0^2 = e_0$, $e_0 \neq 1$, and $ae_0b = (ab)^2 = 1$ shows $e_0 \neq 0$. Set $\nu = b(1-e_0)$. Since $(1-e_0)b = 0$, we have $\nu^2 = 0$, while $a\nu = 1 - e_0 \neq 0$. Thus $(1+\nu)^{-1} = 1 - \nu$, and $(1+\nu)^k = 1 + k\nu$ shows its order is exactly p. If $1 + \nu = \alpha g$ and $g \neq 1$, torsion-freeness makes $1, g, g^2$ distinct, contradicting $(\alpha g - 1)^2 = 0$. The scalar case also contradicts $\nu \neq 0$.

The embedding $\mathbb{F}_p[G] \hookrightarrow K[G]$ is injective on the group basis and preserves every identity and nonidentity. Diagonal matrices with first entries a, b and remaining entries one give matrix failures in every size. Finally sofic group algebras are stably finite by [12, Corollary 4.7 in the preprint]. The scalar failure therefore excludes soficity. These are algebraic consequences of the witnesses, also recorded in [23, Section 2], not additional finite constructions. □

For clarity, the number of relators in the displayed spanning-tree presentation is $|E\Gamma| - |V\Gamma| + |\pi_0\Gamma|$, whereas $|T|/2$ counts its alphabet generators. Neither is the size of an algebraic support, and neither is controlled by the alphabet estimate alone. The final replication and all probability thresholds may depend on the characteristic and on the chosen Hadamard matrix.

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Appendix A. The multiplicity proof of bounded-path exclusion

We prove Theorem 7.7 using the multiplicity argument of [28, Section 4; source file `patterns.tex`], with antecedents in [26]. There are two competing counts. Lemma 7.6 bounds realizations of a fixed labelled image; Lemma 7.5 restricts the words that can satisfy many comparisons. A path may traverse the same edge repeatedly, so occurrences cannot simply be treated as independent edge prescriptions.

The argument proceeds at the level of the injective image. Its boundedly many chains are subdivided on a common approximate grid, allowing repeated occurrences to share word variables. Comparisons then constrain words on linked blocks. Finally, a multiplicity filtration counts each image edge only at the levels at which it occurs; the sum of deterministic first-moment exponents is negative even though no single level is prescribed in advance. The model-specific reservoir and sign bookkeeping are contained in Lemmas 7.6 and 7.5.

Fix K, C, I. All bounded quantities and all uniform $o(L)$ terms in this appendix may depend on this triple and the fixed finite input. Splitting a cyclic comparison at indexing cuts changes the number of comparisons by a bounded amount and preserves the paired positions.

A.1. **The injective image and bounded chains.**

Lemma A.1. *For fixed C, the cycle rank of any finite graph with girth at least L and at most CL edges is bounded independently of L.*

Proof. Delete hanging trees. Circular components contribute at most C. In the remaining graph suppress degree-two vertices and give each resulting edge its positive integer length. The resulting multigraph has minimum degree at least three. Let it have r edges, of total recorded length at most CL.

Start a nonbacktracking walk uniformly on the $2r$ oriented edges and choose each continuation uniformly. The uniform distribution on oriented edges is stationary: an oriented edge from a vertex v has $\deg(v)-1$ possible predecessors, each contributing $1/(\deg(v)-1)$. A walk of k steps therefore has expected recorded length at most kCL/r. At least half its probability is carried by walks of length at most $2kCL/r$. Every individual walk has probability at most $(2r)^{-1}2^{-(k-1)}$, so there are at least $r2^{k-1}$ such walks.

Take $k=\lceil \log_2(8r)\rceil+1$. The lower bound $r2^{k-1}$ for the number of such short walks exceeds the number of ordered endpoint pairs, at most $(2r)^2$. Two distinct short walks share endpoints. Their expansions are distinct reduced paths, whose union contains a cycle: a forest has a unique reduced path between any two vertices. This cycle has length at most $4kCL/r$. Girth gives

$$r \le 4C(\lceil \log_2(8r)\rceil+1),$$

which bounds r in terms of C. Suppression and removal of trees preserve cycle rank. □

Let Δ be the injective image of a putative path system, with at most CL distinct edges. Mark its vertices of degree different from two and all indexed path starts and ends. Its only possible leaves are the endpoints of the one interval. If β is its cycle rank and c its number of components, then

$$\sum_{\deg z\ge 3}(\deg z-2)=2\beta-2c+\#\{z:\deg z=1\}.$$

Lemma A.1 bounds the marked vertices and their total degree. Each component contains a marked path start. Thus Δ consists of boundedly many chains with disjoint degree-two interiors.

Give each chain an orientation. Every indexed path traverses whole chains. The multiplicity μ_e of an undirected image edge is constant along a chain. Two successive uses of the same oriented edge in an immersed indexed path are separated by at least L: the intervening nonempty reduced closed path contains a cycle. Counting both orientations and all paths gives

$$\mu_e \le 2(H/L+K)\le 2(C+K), \qquad \mu_* := \max_e \mu_e \le M_* := \lceil 2(C+K)\rceil . \tag{A.1}$$

An unlabelled pattern records the marked chain graph, orientations, lengths, sides and proper A-colourings, all paths as chain-traversal sequences, the comparison endpoints and directions, and the possible rooted start. All discrete counts other than lengths and endpoints are bounded; those range over $O(L)$ integers. The number of unlabelled patterns is polynomial in L. Nonbipartite A-images have no realization. A colouring costs at most two choices per connected component, hence a bounded factor.

For $1\le j\le \mu_*$ let Δ_j consist of edges of multiplicity at least j and their endpoints. Write E_j, V_j for its edge and vertex counts. Put $k_1=1$ if an interval is present and $k_1=0$ otherwise; put $k_j=0$ for $j>1$.

Lemma A.2 (Multiplicity incidence)**.** *The levels satisfy*

$$\sum_j E_j = H, \qquad \sum_j V_j \le H+k_1, \qquad \sum_j (V_j-E_j-k_j)\le 0. \tag{A.2}$$

Proof. The first identity counts each image edge at its multiplicity levels. For a vertex z, let $a(z)$ count the interval endpoints there, with multiplicity. Each other traversal incidence is paired by an immersed path turn with a different half-edge. For sufficiently large L there are no loops in the image. Hence

$$2\max_{e\ni z}\mu_e \le \sum_{e\ni z}\mu_e + a(z).$$

The vertex occurs at exactly $\max_{e\ni z}\mu_e$ levels. Sum over vertices, use $\sum_z a(z)=2k_1$, and then the first identity, to obtain the other two assertions. The root factor is imposed only at the first level. □

A.2. **A common approximate grid.** Different occurrences of one underlying chain must use the same subdivision, otherwise comparisons do not define constraints on common word variables. Exact alignment may fail because the comparison intervals have different offsets. It suffices to align them up to an $o(L)$ total loss: the ensuing unmatched positions are absorbed by the same error allowance that appears in the bounded-path estimate. The simultaneous approximation below produces such a common grid. Put coordinates $[0, \ell]$ on each oriented chain, with edge positions at half-integers. Split comparisons at chain ends. There are boundedly many resulting maps

$$x \longmapsto \sigma x + a, \qquad \sigma \in \{1, -1\}, \quad a \in \mathbb{Z}.$$

When $\sigma = 1$ the chain-oriented labels agree; when $\sigma = -1$ they are inverse. This includes the original comparison direction and both path traversal orientations.

Let $r_{\mathrm{off}} \geq 1$ bound the number of offsets, padding by zeros. Define

$$Q_L = \left\lfloor L^{1/(2(r_{\mathrm{off}}+1))} \right\rfloor, \qquad R_L = \left\lfloor \sqrt{Q_L} \right\rfloor.$$

Partition $[0,1)^{r_{\mathrm{off}}}$ into $Q_L^{r_{\mathrm{off}}}$ boxes. Two of the vectors of fractional parts $\left(\{\ell a/L\}\right)_a$, for $0 \leq \ell \leq Q_L^{r_{\mathrm{off}}}$, lie in the same box. Their difference gives an integer $1 \leq u \leq Q_L^{r_{\mathrm{off}}}$ with $\operatorname{dist}(ua/L, \mathbb{Z}) \leq 1/Q_L$ for every offset. Choose the least such u and put

$$s = \frac{L}{uR_L}. \tag{A.3}$$

Uniformly over patterns,

$$s \geq L^{1/2+1/(4(r_{\mathrm{off}}+1))}, \qquad s \leq L/R_L = o(L), \qquad \operatorname{dist}(a/s, \mathbb{Z}) \leq R_L/Q_L = o(1).$$

Use as blocks the edge positions in full cells $[ks, (k+1)s)$ lying within a chain. Each block i has $h_i = s + O(1)$ positions and inherits a multiplicity μ_i. The total underlying end-fragment length is $O(s)$; the occurrence length is also $O(s)$ by (A.1).

Choose a uniform $\kappa_L \to 0$ larger than a suitable fixed multiple of $R_L/Q_L + L^{-1/2}$. Link two full-block occurrences when one subcomparison identifies substrings of length at least $(1-\kappa_L)s$, and retain the entire substring constraint. The overlap exceeds half a block for large L. Disjointness of compared positions therefore makes these links a partial matching of block occurrences.

Lemma A.3. *If b_* is the number of unmatched full-block occurrences, then*

$$sb_* \leq b_0 + O(s + L/s) = b_0 + o(L). \tag{A.4}$$

Proof. Only boundedly many block occurrences lie within $3s + O(1)$ of a subcomparison endpoint or a chain end. Every other occurrence is wholly unpaired or wholly inside a single subcomparison. In the second case the grid displacement is at most $(R_L/Q_L)s$ and its partner cell is full. Reflection changes endpoint conventions by at most two positions. Thus the overlap exceeds $(1-\kappa_L)s$ and the occurrence is linked. Every other unmatched occurrence contributes at least $s-1$ unpaired positions. There are $O(L/s)$ occurrences, giving (A.4). □

Forget occurrence indices, retaining all links, to obtain a multigraph on underlying blocks. A link can be a self-link. For a link component U, including isolated blocks, write

$$S_U = \sum_{i \in U} \mu_i, \qquad z_U = \max_{i \in U} \mu_i, \qquad b_U = \#\{\text{unmatched occurrences in } U\}.$$

Then $\sum_U b_U = b_*$. The following deterministic estimate controls its word freedom.

Lemma A.4. *There is a uniform $\xi_L \to 0$ such that:*

(i) *after the word on one block of U is fixed, the number of assignments to its other blocks satisfying all links is at most $\exp(\xi_L s|U|)$;*
(ii) *if U has a self-link, the same bound holds without fixing any initial word;*
(iii) *if U has no self-link, $2z_U \leq S_U + b_U$.*

Proof. For (i), propagate along a spanning tree of U. A known block determines at least $(1-\kappa_L)s$ positions on the next one, leaving at most $\kappa_L s + O(1)$ freely chosen letters. Ignoring additional constraints only enlarges the bound.

For (ii), begin at a self-linked block. A translation constraint has nonzero displacement a, because a zero displacement would match occurrences of the same underlying edge, which is forbidden. Its equality graph joins positions x and $x + a$ over a consecutive domain of at least $(1 - \kappa_L)s$ positions. It is a forest, separately a subgraph of a line in each residue class modulo $|a|$, and has at most $\kappa_L s + O(1)$ components. It has only $|T|^{\kappa_L s+O(1)}$ possible initial words. A reflection self-link is impossible: the domain and image have a nonempty reflection-invariant interval of overlap, whose center forces either a self-inverse letter or two adjacent inverse letters. Both contradict a reduced word over a fixed-point-free alphabet. Now propagate as in (i).

For (iii), a block of multiplicity z_U has at least $z_U - b_U$ matched occurrences. Without self-links their partners are distinct occurrences of other blocks, of which there are $S_U - z_U$. Rearrangement proves the assertion. The estimates hold with ξ_L a sufficiently large fixed multiple of $(\kappa_L + L^{-1/2}) \log |T|$. □

A.3. **Bins and projections of complete assignments.** We group words according to their weight so that squared-word decay becomes a bound on the number of choices in each group. A comparison may determine the word on a block that is absent at one particular multiplicity level. Counting solutions to only the comparisons visible at that level would therefore lose information. We first impose *all* comparisons on complete assignments and only then project these assignments to each level. This preserves the constraints propagated from blocks outside the level. For each chain-oriented block word w_i record the bin

$$d_i = \lfloor -\log P_\Theta(w_i) \rfloor .$$

By Lemma 7.5, for all sufficiently large L and uniformly in the patterns,

$$\gamma s/2 \le d_i \le a_0 s, \qquad \#\{w \in T^{h_i} : \lfloor -\log P_\Theta(w) \rfloor = d_i\} \le \exp(2d_i - \gamma s). \tag{A.5}$$

Indeed any individual word has weight at most $\exp(-\gamma h_i)$, giving the first lower bound after taking the floor. A word in its bin has weight greater than $\exp(-d_i - 1)$; the squared-word sum gives at most $\exp(2d_i + 2 - 2\gamma h_i)$ words. The upper bound on d_i uses $h_i - 1 \le s$ and the definition of a_0.

There are $O(L/s)$ blocks and $O(s)$ bins per block. The number of enlarged patterns, including the polynomial choice of unlabelled pattern, is therefore

$$\exp\big(O((L/s)\log(s+2)) + O(\log L)\big) = \exp(o(L)). \tag{A.6}$$

The grid and links were deterministic functions of the unlabelled pattern, so introduce no additional choices.

Fix an enlarged pattern. Let $\mathcal{A}$ be the finite set of complete edge-label assignments on its abstract image satisfying immersion, every comparison, and all bins. Define

$$\mathcal{A}_j = \{\alpha|_{\Delta_j} : \alpha \in \mathcal{A}\}.$$

These are restrictions of complete compatible assignments, not all assignments satisfying only the constraints visible at level j. Let Z_j count their injective realizations, imposing the prescribed root just when $k_j = 1$. Any realization of the full pattern gives $Z_j \ge 1$ at every level.

For one member of $\mathcal{A}_j$, Lemma 7.6 applies. The active chains are whole chains of the original image, so their number is bounded even if Δ_j is disconnected. Discarding all factors except turns internal to active full blocks increases the bound, since $0 \le \Theta < 1$. Absorbing f^{-Q} gives expectation at most

$$n^{V_j - E_j - k_j} \exp\left(- \sum_{i:\mu_i \ge j} d_i + o(L) \right). \tag{A.7}$$

The number of restricted assignments is at most

$$|\mathcal{A}_j| \le \exp\left(o(L) + \sum_{\substack{U \text{ without self-link} \\ z_U \ge j}} (2 \min_{i \in U} d_i - \gamma s) \right). \tag{A.8}$$

For a component without a self-link, select a block with least bin in the *whole* component. It may be absent from this level. Equation (A.5) counts that word, and Lemma A.4(i) counts its extensions. Projection cannot increase the number of complete assignments. A component with a self-link needs no anchor, by part (ii). Propagation contributes $\xi_L s\, O(L/s) = o(L)$, and end fragments contribute $|T|^{O(s)} = \exp(o(L))$. Junction constraints and unused comparisons only reduce the count.

Let B_j be the product of the right sides of (A.7) and (A.8), with common uniform error bounds. First moments give

$$\mathbb{E}_L Z_j \le B_j, \qquad \mathbb{P}_L(\text{full pattern occurs}) \le \min_j B_j. \tag{A.9}$$

The B_j are deterministic bounds. No independence of the events $\{Z_j \ge 1\}$ is asserted or needed.

A.4. **Summing over multiplicity levels.** An occurrence of the full pattern produces an image at every level, so any one level with a small first moment suffices. We find such a level by summing the logarithms of the deterministic bounds. The multiplicity-incidence inequality makes the total vertex exponent nonpositive, leaving the loss in word freedom to outweigh the permitted unmatched positions. Sum $\log B_j$ over j. The coefficient of $\log n$ is nonpositive by Lemma A.2. For a component U without a self-link, put $d_{\min} = \min_{i \in U} d_i$. Its remaining contribution is at most

$$z_U(2d_{\min} - \gamma s) - \sum_{i \in U} \mu_i d_i \le (2z_U - S_U) d_{\min} - \gamma s z_U.$$

If $2z_U \le S_U$, use $d_{\min} \ge \gamma s/2$ to get at most $-\gamma s S_U/2$. Otherwise use $d_{\min} \le a_0 s$ and $2z_U - S_U \le b_U$ to get at most $a_0 s b_U - \gamma s S_U/2$. A self-linked component has no initial-word term, and contributes at most $-\gamma s S_U/2$ directly.

Full blocks account for $H - O(s)$ occurrences. Replacing h_i by s costs $O(L/s)$, since multiplicity is bounded. Therefore

$$s \sum_U S_U = H + o(L), \qquad s \sum_U b_U \le b_0 + o(L).$$

There are at most M_* levels, so their uniform errors still sum to $o(L)$. We obtain

$$\sum_{j=1}^{\mu_*} \log B_j \le -\frac{\gamma}{2} H + a_0 b_0 + o(L). \tag{A.10}$$

Our $\varepsilon = \gamma/[32(1 + a_0)]$ is fixed independently of K, C, I. If $b_0 \le \varepsilon H$ and $H \ge L$, the right side is at most $-\gamma H/4$ for all sufficiently large L, uniformly over enlarged patterns. Some level satisfies

$$B_j \le \exp(-\gamma H/(4\mu_*)) \le \exp(-\gamma H/(4M_*)).$$

Combine (A.9) with (A.6). The total forbidden-system probability is at most

$$\exp(-\gamma L/(4M_*) + o_{K,C,I}(L)) \longrightarrow 0.$$

This proves Theorem 7.7, with its stated order of quantifiers.

The restriction to distinct underlying edges is essential in the self-link argument. The prescribed start of the possible interval is likewise essential in (A.2): an unrooted interval would leave an uncompensated positive vertex exponent. The proof establishes neither a bound for arbitrary unrooted intervals nor a bound simultaneous over an unbounded number of path-system parameters.

Appendix B. The cone criterion and its geometric hypotheses

We give the proof of Theorem 8.1 following Mghirbi–Douglas [23, Sections 3–4]. The same picture estimate has three consequences, according to what is allowed on the outside boundary. An essential sphere has no exterior boundary, so excluding its picture proves asphericity. A rooted picture has a specified path with fixed endpoints, so excluding it protects the root label. A free-boundary picture represents an arbitrary null word; bounding its inner cone length by the exterior length gives the linear area inequality and hence hyperbolicity.

These are distinct geometric problems, not interchangeable versions of a spherical argument. Proposition B.4 treats them through one estimate: short closures produce graph paths, interval pairings organize their boundary identifications, and planar separators isolate bounded forbidden configurations. Minimality ensures that each comparison uses distinct underlying graph edges.

The short-closure, interval, separator and cone-picture arguments come from [29, Sections 4–5] and [28, Sections 5–6]. The free-boundary application to these cone complexes is Mghirbi–Douglas's; related accounting for another complex appears in [27, Theorem 3.1 and Lemma 3.8]. For the classical relation between diagrams, pictures, and identities among relations, see [20, Chapter V] and [7, 2]. The separator input is [18].

Fix a graph satisfying the pattern-free conditions preceding Theorem 8.1. In this appendix write $D_c, U_c, \eta_c, K_c, C_c, I_c$ for the six constants in (8.1). They satisfy

$$\frac{3D_c}{U_c} \le \frac{\varepsilon}{32}, \qquad 2\eta_c U_c \le \frac{1}{16}, \qquad 2\eta_c U_c \le \frac{\varepsilon}{32}, \qquad \frac{C_{\mathrm{sep}}}{\sqrt{K_c}} \le \eta_c. \tag{B.1}$$

B.1. **Short closures, separators, and intervals.** Long boundary paths are first closed by short additions, pairing arcs are gathered into intervals, and planar separation breaks the picture into bounded clusters. The losses at these three operations are exactly what the constants in (B.1) control.

Lemma B.1. *Every nonempty immersed path in Γ can be extended to a cyclically immersed closed path by adding at most D_cL traversals, without cancelling an original traversal.*

Proof. At a vertex z forbid at most two incident undirected edges. Deleting them leaves minimum degree at least three, since the original minimum is five and the girth is at least three. Set $k = \lceil \log_2 |V\Gamma| \rceil + 1$. There are at least $3 \cdot 2^{k-1} > |V\Gamma|$ nonbacktracking walks of length k from z. Two distinct such walks share an endpoint. Reducing their concatenation with one reversed removes only a common terminal segment, and leaves a nonempty reduced based loop of length at most $2k$, with first and last edges avoiding the forbidden edges.

For a path from a to b, choose a geodesic R from b to a, of length at most dL. When R is nonempty, insert a based loop at b avoiding the incoming path edge and first edge of R, and one at a avoiding the last edge of R and first edge of the path. The resulting concatenation is cyclically immersed. If R is empty, one avoiding loop suffices. The added length is at most

$$dL + 4k \le dL + 4(\lambda_{\mathrm{geo}} L + 2) \le D_c L.$$

The auxiliary based loops need not be cyclically immersed individually; the indicated joins are the ones used. □

Lemma B.2. *A planar multigraph with s vertices has a vertex set of size at most $C_{\mathrm{sep}} s/\sqrt{K}$ whose deletion leaves components of size at most K.*

Proof. Discard loops and repeated edges. By Lipton–Tarjan [18, Corollary 2], a component of size a has a separator of size at most $2\sqrt{2a}$ leaving parts of size at most $2a/3$. Apply this recursively to components larger than K, charging $2\sqrt{2}/\sqrt{a}$ to each of their vertices. Along the successive calls involving a vertex, component sizes decrease by at least a factor $2/3$. Summing backward from the last such size, which exceeds K, bounds the total charge by

$$\frac{2\sqrt{2}}{\sqrt{K}} \sum_{j \ge 0} (2/3)^{j/2} = \frac{C_{\mathrm{sep}}}{\sqrt{K}}.$$

Summing over all vertices proves the estimate. □

Lemma B.3 (The interval lemma)**.** *Let a sphere contain disjoint closed discs $D_1, \dots, D_{N_0}$ whose oriented boundaries carry nonempty cyclic words. Assume they are cyclically reduced except possibly at one marked break. Disjoint arcs in their complement pair every letter occurrence with an inverse letter. Divide the boundaries into $N' \geq N_0$ nonempty consecutive pieces, cutting at the break. Then the arcs are described by at most $6N'$ pairs of intervals, each within one piece and paired in reversed order.*

Proof. A regular neighborhood of discs and arcs has Euler characteristic $N_0 - E$, where E is the number of arcs. For each complementary region Q, let ℓ_Q count incident arc sides, equivalently boundary gaps. Additivity and double counting give

$$\sum_Q (\ell_Q - 2\chi(Q)) = 2N_0 - 4. \tag{B.2}$$

No simple connectedness of complementary regions is assumed. Call a gap regular if it belongs to a disc-shaped digon whose sides are two distinct arcs. Its opposite gap is its partner. A monogon joins neighboring inverse letters and can occur only at the one break, so there are at most $M = 1$ monogons. A digon using both sides of one arc has degree-one boundary gaps; there are at most $2N_0$ such gaps. Every other region is a disc with at least three sides, or a planar surface with at least two boundary components; in either case

$$\ell_Q \leq 3(\ell_Q - 2\chi(Q)).$$

Equation (B.2) therefore bounds the number of nonregular gaps by

$$M + 2N_0 + 3(2N_0 - 4 + M) \leq 8N_0.$$

Cut these gaps, the piece boundaries, and one indexing gap on each original circle, always using the marked break when present. If an additional cut gap is regular, also cut its partner. There are at most $8N_0 + 2(N_0 + N') \leq 12N'$ cuts. The cut set meets every circle and is stable under regular partners. Between cuts, following successive digons pairs an entire run with a run in reversed order. A run cannot pair with itself: its middle would be a self-paired letter or two adjacent inverse letters, and the break was cut. Thus runs pair in at most $6N'$ pairs. □

The lemma concerns words, not necessarily paths in Γ. In particular it applies to the exterior word of an arbitrary van Kampen disc.

B.2. **Pictures over abstract cones.** To use classical pictures, we first replace each cone by relator discs for a free basis of its attaching graph. The replacement is a homotopy equivalence relative to the rose. It does not require the image of an attached cone to embed in the quotient complex; fillings and surgeries are carried out in the specified abstract cone. Choose a spanning tree in a component Λ of Γ. Attach discs to Λ along the corresponding free-basis loops for its nontree edges, obtaining Y_Λ. Collapsing the tree identifies Y_Λ with a contractible bouquet of circles individually filled by discs. Both Y_Λ and the abstract cone on Λ are contractible CW complexes containing Λ as a subcomplex. Identity on Λ extends to maps between them, whose composites are homotopic to identity relative to Λ by cellwise extension. Replacing the attached cone by Y_Λ therefore gives a homotopy equivalent complex relative to the rose.

This reduction gives the usual picture representation: a map of a sphere, or a null-homotopy with prescribed boundary, is represented by finitely many disjoint inner discs, each factoring through one abstract cone with boundary a closed graph path, and disjoint complementary arcs pairing inverse letters. One can obtain it directly from a cellular approximation in the relator complex by taking transverse inverse images of points in relator discs and rose edges; see [2] and [29, Lemma 5.2]. Closed components of an inverse image of a rose-edge point carry no boundary letters. In the spherical case collapsing one splits the homotopy class into a sum of translates of two spherical classes; if the original class is essential, at least one is essential. Repetition removes all such components from an essential representative. The same relative construction gives disc pictures.

For a free picture the sphere has an additional exterior disc reading the inverse of a fixed word w; its interior is formal. For a rooted picture it instead reads a specified graph path from a root,

with its starting gap marked. The cone length is $H_0 = \sum_i |\gamma_i|$, summed over inner boundaries. Every closed graph path of length h is, in the free group on the rose, a product of at most h conjugates of the fundamental-cycle relators: insert tree paths at each traversed edge and cancel successive tree paths. Therefore a free picture gives

$$\mathrm{Area}(w) \le H_0. \tag{B.3}$$

Consider an essential spherical picture; a picture for a fixed nonempty cyclically reduced free word; or a rooted picture with prescribed, distinct graph endpoints. Minimize total graph-boundary length, then the number of inner discs. In the free case only inner length is minimized. Empty inner discs are deleted, and backtracking is tightened, never across the rooted break. Tightening preserves graph endpoints and is a homotopy in the same abstract cone. Minimality consequently makes inner boundaries nonempty and cyclically immersed, and a rooted outer path nonempty and immersed.

A minimal picture cannot have an arc joining two graph-boundary occurrences of the same undirected edge. Such occurrences traverse it in opposite directions. A narrow band along the arc lifts to that same abstract edge, and the following surgeries delete the two occurrences.

Two inner discs. Boundaries ep and $e^{-1}q$ merge to one boundary pq. The two discs and band factor through the same contractible cone, so replacement preserves the homotopy class relative to the complement.

An inner disc and a rooted exterior. The replacement $u_1eu_2 \mapsto u_1qu_2$, using an inner boundary $e^{-1}q$, preserves the graph endpoints and the root break, while removing that inner disc.

Two occurrences on the rooted exterior. Write its linear boundary as $u_1eu_2e^{-1}u_3$. The exterior and band form an annulus. Retain the complementary disc containing the marked break; its boundary is u_1u_3 with the same endpoints. The remaining original picture supplies its null-homotopy. No filling of the discarded path is assumed.

Two occurrences on one inner disc. A boundary $epe^{-1}q$ splits into the two closed paths p, q. Cap both in the same abstract cone. For a rooted or free picture, retain the capped side containing the exterior. For an essential sphere, lift to the universal cover and use one lift of that contractible cone for both caps. In second homology the original class is the sum of the two capped classes, because their difference factors through the cone. Hurewicz identifies its nonzero spherical class with nonzero homology; at least one capped sphere remains essential.

Every move decreases length by at least two, or removes an empty disc; subsequent tightening does not increase it. In the free case no move touches the exterior word. Thus minimal pictures are reduced: all graph paths have the stated immersion properties, and each arc between graph boundaries joins distinct underlying edges. There is at least one inner disc. Otherwise an essential sphere maps into the rose, or a nonempty reduced free/rooted word is trivial in a free group, both impossible. These arguments are in the abstract cones before passage to their quotient and require no embedding of a cone in X.

B.3. **One estimate for spheres and both disc problems.** The free boundary contributes at most its length in unpaired graph occurrences. Closures and separator deletions contribute the other losses, controlled by the chosen constants. If the exterior were too short, a surviving cluster would violate pattern-freeness. With no free exterior this excludes the spherical and rooted pictures outright.

Proposition B.4. *A reduced rooted picture cannot exist. For a reduced free or spherical picture, with exterior length ℓ_∞ (zero for a sphere),*

$$H_0 \le \frac{16}{13\varepsilon}\ell_\infty.$$

Proof. Let H be the total graph-boundary length, so $H = H_0$ in the free and spherical cases and $H = H_0 + |u|$ in the rooted case. There is an inner disc, hence $H_0 \ge L$. Suppose the picture is rooted, or $\ell_\infty < 13\varepsilon H_0/16$.

Keep every graph boundary of length at most U_cL. Split a longer boundary of length h into $\lceil h/(U_cL)\rceil$ consecutive segments of almost equal lengths, in linear order for a rooted path. Since

$U_cL \geq 6$, their lengths lie between $U_cL/3$ and U_cL. Close each by Lemma B.1, declaring its added positions unpaired. The number of pieces N' and added length satisfy

$$H_{\mathrm{add}} \leq (2D_c/U_c)H, \qquad N'L \leq H + L \leq 2H. \tag{B.4}$$

All pieces are cyclically immersed and have at least L original occurrences, except possibly the intact rooted path.

Apply Lemma B.3 to the original picture cut into these pieces; in the free case the exterior is one additional piece. There are at most $12H/L$ interval pairs. Indeed without a rooted path $N'L \leq H$ and the possible extra piece contributes at most one more, while $H \geq L$; in the rooted case $N'L \leq 2H$. Discard all pairs touching the free exterior; they cost at most ℓ_∞ graph occurrences.

Make a planar multigraph with one vertex per graph piece and edges the old pairing arcs between them. Planarity follows by placing piece vertices in disjoint sectors inside their original discs. Closures need not be embedded: they carry no arcs. Lemma B.2 deletes at most $\eta_c N'$ pieces, leaving clusters of at most K_c pieces. The deleted pieces carry at most $2\eta_c U_c H$ original occurrences. Each surviving occurrence whose partner was deleted is charged to one such deleted occurrence. Because both ends of an interval pair lie in single pieces, surviving pairs stay whole and lie within one cluster.

Let H_Z, b_Z, I_Z be the total length, unpaired length and number of retained interval pairs in a cluster. Equations (B.1)–(B.4) give

$$\begin{aligned}
\sum_Z H_Z &\geq (1 - 2\eta_c U_c)H \geq 15H/16,\\
\sum_Z b_Z &\leq (2D_c/U_c + 2\eta_c U_c)H + \ell_\infty \leq \varepsilon H/16 + \ell_\infty,\\
\sum_Z I_Z &\leq 12H/L.
\end{aligned}$$

Clusters with $b_Z > \varepsilon H_Z$ account for total length at most $H/16 + \ell_\infty/\varepsilon$; those with $I_Z > 192H_Z/L$ account for at most $H/16$. The other, good clusters therefore carry total length at least

$$13H/16 - \ell_\infty/\varepsilon.$$

This is positive in the free or spherical case by the assumed inequality and $H = H_0$. Every cluster then has length at least L. In the rooted case $\ell_\infty = 0$; a cluster shorter than L can only be the intact rooted path alone, of length $h < L$. As an inner disc is present, $H \geq h + L > 2h$. Discarding this possible cluster costs less than $H/2$, leaving a good cluster of length at least L.

In every case some good cluster satisfies

$$L \leq H_Z \leq K_c(U_c + D_c)L = C_cL, \quad b_Z \leq \varepsilon H_Z, \quad I_Z \leq 192H_Z/L \leq I_c.$$

It has at most K_c paths, with exactly the allowed closed/rooted possibilities. Its compared positions are disjoint and use distinct underlying edges. It is a forbidden pattern, a contradiction. □

Completion of Theorem 8.1. If $\pi_2(X) \neq 0$, an essential spherical picture of least complexity is reduced, contradicting Proposition B.4 with $\ell_\infty = 0$. Thus $\pi_2(X) = 0$. The simply connected universal cover is two-dimensional and acyclic, by Hurewicz and dimension. A first nonzero homotopy group would contradict Hurewicz, and Whitehead then makes it contractible [15, Theorems 4.32 and 4.5]. Hence X is aspherical.

If a path from a designated root to a different vertex had trivial label, a minimal rooted picture for those fixed endpoints would be reduced, which Proposition B.4 forbids. This proves root protection. For a free word, free and cyclic reductions do not increase length and conjugation preserves area. Its minimal capped picture gives (8.2) by (B.3) and the same proposition.

The finite complex gives a finite presentation. To exclude torsion, restrict the augmented free cellular $\mathbb{Z}G$-resolution of length two to a supposed cyclic subgroup C_ℓ of prime order. It remains free over $\mathbb{Z}C_\ell$, so it would give $H^j(C_\ell; \mathbb{F}_\ell) = 0$ for $j > 2$. But the cyclic periodic resolution alternates $u - 1$ and $1 + u + \cdots + u^{\ell-1}$; after applying $\mathrm{Hom}_{\mathbb{Z}C_\ell}(-, \mathbb{F}_\ell)$, every differential is zero.

Thus $H^j(C_\ell;\mathbb{F}_\ell)=\mathbb{F}_\ell$ in every degree, a contradiction; compare [5, Chapter VIII]. Finally the linear area bound implies word-hyperbolicity by [4, Theorem III.H.2.9]. □

Appendix C. Finite specifications and exact verification

The finite constructions can be reconstructed without the final graph matching. We give the characteristic-three difference set and the formulas for the small Paley matrices, followed by a polynomial certificate for the restricted comparison with Mghirbi–Douglas's unchanged kernel. The last item is a mathematical obstruction, not merely a test of the implementation. The supplementary files `SOURCES.md` and `DEPENDENCIES.md` identify the source versions and the mathematical inputs used in each theorem. The verification code uses Python integers and rational arithmetic for certificates, and fixed-width integer arrays only for the explicitly bounded small incidence and Hadamard calculations.

C.1. **The characteristic-three difference set.** The ordered set $\mathcal{D}\subset\mathbb{Z}/6643$ used in Section 6 is

0	1	3	9	27	32	81	96	151	199	243	288
453	457	509	584	597	614	729	803	864	904	932	1003
1133	1359	1371	1493	1527	1745	1752	1791	1842	1856	2044	2187
2225	2384	2409	2419	2482	2525	2592	2712	2796	2833	2956	3009
3056	3233	3292	3399	3418	3478	3554	3611	3791	3802	4019	4077
4113	4190	4429	4479	4495	4581	4730	4763	5110	5235	5256	5373
5414	5526	5568	5588	5696	5905	5927	6132	6397	6561		

All $82\cdot 81=6642$ ordered nonzero differences occur exactly once. This is a direct finite certificate for the projective plane used here, independently of the general Singer existence theorem. The two pencils through $0,1$ have line index sets $-\mathcal{D}$ and $1-\mathcal{D}$, respectively. Their unique common index is 0. The assignment rule in Lemma 6.1, together with increasing cyclic indices, determines all 111 nonzero coefficients and every repair array. Thus no longer coefficient list is needed to specify the input.

The verifier reconstructs these arrays from this set rather than assuming a stored inverse permutation or stored repairs. It checks the difference counter, all eleven auxiliary correlations, the local pencil identity, every residue, the complete involution, both integer repairs, every inverse pool equality, reservoir slots, class offsets, positivity, and the exact rational spectral margin. The file `data/characteristic_three.json` stores the resulting arrays for comparison with that reconstruction.

C.2. **Paley matrices used in Table 1.** These formulas specify the actual auxiliary matrices used in the small examples. Normalizing them recovers both design Gram identities of Lemma 2.2, so the table depends on finite matrix arithmetic rather than an external list of known orders. Let χ be the quadratic character of a finite field $\mathbb{F}_s$, with $\chi(0)=0$, and put $C_{ij}=\chi(i-j)$. The standard character identity gives $CC^{\mathsf{T}}=sI-J$ and $C\mathbf{1}=0$. If $s\equiv 3 \pmod 4$, the matrix

$$H=\begin{pmatrix}1 & \mathbf{1}^{\mathsf{T}}\\ -\mathbf{1} & C+I\end{pmatrix}$$

has $HH^{\mathsf{T}}=(s+1)I$. If $s\equiv 1 \pmod 4$, put

$$A_s=\begin{pmatrix}0 & \mathbf{1}^{\mathsf{T}}\\ \mathbf{1} & C\end{pmatrix},\qquad H=\begin{pmatrix}A_s+I & A_s-I\\ A_s-I & -A_s-I\end{pmatrix}.$$

Here $A_s^2=sI$, so $HH^{\mathsf{T}}=2(s+1)I$. These are Paley's type I and type II constructions [30]. Row and column sign changes normalize the matrices without changing their Hadamard identities.

The implementation constructs prime fields directly and uses $\mathbb{F}_{25}=\mathbb{F}_5[u]/(u^2-2)$ for the order-52 example. The polynomial is irreducible because 2 is a nonsquare modulo five. Every entry and both design Gram identities are checked after normalization. The six resulting matrices are stored in `data/hadamard_matrices.json`. No external matrix database or third-party matrix-generation code is used.

C.3. **A spectral obstruction for the unchanged MD model.** The following comparison concerns only the specific auxiliary blocks, frequencies, and conditional kernel of [23, Section 7]. It does not prohibit smaller parameters after a change of finite design.

Proposition C.1. *Keep the MD direct-finiteness auxiliary system, with $e = p^2 + p + 1$ and auxiliary reduced-turn weight $(p-1)/p$, but replace its plane order by $q = p^4$. Its squared-turn kernel has spectral radius greater than one for every odd prime p. In fact the same quotient obstruction holds for every real $p \geq 3$.*

Proof. Set $v = q^2 + q + 1$, $a = (q+1)^{-2}$, $b = (q+1)^2/v^2$, and $g = (p-1)/p$. The three-class quotient is

$$Q = \begin{pmatrix} eb & (e-1)g^2 & (v-e)b \\ (e-1)a & eb & (v-e)a \\ ea & eb & (v-e-1)a \end{pmatrix}.$$

Direct rational expansion at $q = p^4$ gives

$$\det(I - Q) = \frac{-P(p)}{(p^4+1)^3(p^2-p+1)^4(p^2+p+1)^2(p^4-p^2+1)^4}, \tag{C.1}$$

where the coefficients of P, in descending order from degree 31 to degree zero, are

$$(1, 1, 0, -1, 1, 12, -4, -14, 11, 31, -21, -31, 26, 43, -37, -46,$$
$$40, 38, -45, -36, 37, 18, -31, -17, 22, 4, -15, -1, 7, -2, -2, 1).$$

All 32 coefficients of $P(u+3)$ are positive; its constant is 833193976801090. This is an integer certificate: if c_k denotes the coefficient of p^k in the displayed list, each coefficient is the explicitly checkable sum

$$[u^j]P(u+3) = \sum_{k=j}^{31} c_k \binom{k}{j} 3^{k-j} > 0 \qquad (0 \leq j \leq 31).$$

The accompanying program expands the determinant independently, compares every coefficient of P, and verifies these 32 positive sums. Thus (C.1) is negative for $p \geq 3$. The real polynomial $\det(tI - Q)$ is positive for sufficiently large positive t, so it has a root greater than one. That root is an eigenvalue of the full equitable kernel. Its spectral radius is therefore greater than one. □

The observation following [23, Lemma 7.8] shows that their displayed sufficient inequality fails at $q = p^4$. Proposition C.1 instead verifies failure of contraction for the unchanged kernel itself. Neither statement claims optimality over other auxiliary designs or other frequency choices.

C.4. **Reproduction and scope.** From the supplementary package root run

```
python -m pip install -r requirements.txt
python verify/verify.py --output checks/results.json
```

Assertions must remain enabled. The program reconstructs the finite data, compares the two stored certificate files, and writes a new results file; it does not modify the input certificates. It also checks the rational certificates of Table 1, the polynomial in (C.1), the error constants in (6.5), the degree-based girth parameters, and sample boundary values for both asymptotic parameter deductions. The universal inequalities in Section 5 are proved there, not inferred from these samples.

The large Hadamard matrices supplied by the external existence theorems are not constructed by these programs, and the complementary-sequence computations underlying those sources are not repeated. The programs check finite arithmetic and specified finite arrays; they neither find the terminal graph matching nor decide equality in its group. The probabilistic and topological conclusions are supplied by their separately stated proofs and cited inputs.

Appendix D. Two support restrictions

The alphabet estimates in the main text count oriented labels, whereas algebraic support counts distinct group elements after coefficients have been collected. This appendix records two restrictions on the latter quantity. An elementary coefficient bound gives a support-height restriction; fixing all finite equality patterns then gives an effective, but not practical, characteristic cutoff.

The finite-pattern viewpoint is developed for left-invertible elements and universal groups by Dykema–Heister–Juschenko [11, Section 4], and for small-support zero divisors by Schweitzer [31, Section 2]. Fixed-support coefficient equations also appear in Gardam's work [13, Sections 1 and 3]. The Nullstellensatz reduction to finite fields is recorded in [3, Section 1]. For the transfer to characteristic zero, Ivanov [16, Section 8.3] fixes the group and the supports; Mghirbi–Douglas [23, Proposition 8.3] allow the group to vary with a fixed coincidence pattern. Here characteristic-zero direct finiteness supplies the obstruction for one-sided inverses.

Proposition D.1 (Support and coefficient height)**.** *Let $A, B \in \mathbb{Z}[G]$ have distinct support sizes s, t, and coefficient absolute values bounded by H_A, H_B. Suppose reduction modulo a prime p preserves both supports, giving $a, b \in \mathbb{F}_p[G]$. If*

$$p > \min(s,t) H_A H_B,$$

then $ab = 0$ implies $AB = 0$, and $ab = 1$ implies $ba = 1$. Consequently a direct-finiteness failure with these lifts requires $p \le \min(s,t) H_A H_B$. In particular, when both collected coefficient lists consist of signs, both supports have size at least p.

Proof. A product fiber $\{(i,j) : x_i y_j = g\}$ has at most $\min(s,t)$ pairs: fixing either index determines the other. Each coefficient of AB consequently has absolute value at most $M = \min(s,t) H_A H_B$. If $ab = 0$, each is divisible by $p > M$ and hence is zero. If $ab = 1$, all nonidentity coefficients vanish integrally, and $AB = k\mathbf{1}$ for an integer $k \equiv 1 \pmod p$, necessarily nonzero. Characteristic-zero direct finiteness applied to A and B/k gives $BA = k\mathbf{1}$, and reduction gives $ba = 1$. Here we use Kaplansky's theorem in the form proved by Montgomery [25]. It is not necessary to force $k = 1$. □

A bound on coefficient height is unnecessary if only an effective cutoff is sought. With bounded support sizes there are finitely many patterns of equality among the product terms. For each realizable pattern, characteristic-zero direct finiteness rules out the coefficient equations for a failure, and a rational algebraic certificate excludes all sufficiently large characteristics.

Proposition D.2 (A finite-characteristic cutoff at bounded supports)**.** *For each pair of positive integers s, t, there is an effectively computable integer $B(s,t)$ such that, for every group G and every field K of prime characteristic $p > B(s,t)$,*

$$|\operatorname{supp} a| \le s, \quad |\operatorname{supp} b| \le t, \quad ab = 1 \text{ in } K[G] \quad \Longrightarrow \quad ba = 1.$$

Proof. For fixed support sizes enumerate all formal equality patterns on $x_i y_j$, $y_j x_i$, and 1, together with the requirements that the x_i are distinct and the y_j are distinct. There are finitely many. For each pattern choose in turn a block on which a coefficient of $ba - 1$ is required to be nonzero. The equations $ab = 1$, nonvanishing of every support coefficient, and nonvanishing of that defect form a finite system over $\mathbb{Z}$; nonvanishing is encoded by multiplying by an auxiliary inverse variable.

For a group pattern which is realizable, that system has no complex solution, by characteristic-zero direct finiteness. The weak Nullstellensatz puts 1 in its coefficient ideal over $\mathbb{Q}$: a certificate over $\mathbb{C}$ is a finite linear system with rational coefficients and hence has a rational solution. Clearing denominators gives a nonzero integer d in the ideal over $\mathbb{Z}$. Solutions are impossible in characteristics not dividing d. Taking a bound at least 2 and at least every such $|d|$ for the finitely many realizable patterns proves existence of the cutoff.

For effectivity, present the universal group imposing each pattern's equalities. The pattern is realizable precisely when all required inequalities persist in that group. Run in parallel an enumeration of relation proofs that one required inequality collapses, and an enumeration

of rational polynomial identities putting 1 in the coefficient ideal. An unrealizable pattern terminates by the first search; a realizable one terminates by the second. The finite collection of tasks all terminate. Whenever the second search terminates, record its cleared denominator d; the maximum of their absolute values and 2 is a computable bound. This procedure does not decide arbitrary word problems: it uses two semidecision searches, with termination guaranteed here by the alternative algebraic certificate. Repeat for all support sizes up to s, t. □

The proof is a bounded-support consequence of the same finite-pattern transfer method, not a claimed new general mechanism. No useful numerical value of $B(s, t)$ is supplied. Bounded-support *zero divisors* remain different: Proposition 8.3 of [23] transfers such examples in infinitely many characteristics to characteristic zero, but neither the finite design nor the support estimates above provide them.

In (8.6), signs are attached to graph vertices before collection in G. Root protection identifies only the identity fiber. It does not imply that all nonroot labels are distinct, or that the collected coefficients are signs. Thus Proposition D.1 must not be applied to those vertex lists as though they were already distinct supports. Likewise a growing number of graph vertices, by itself, does not prove growing algebraic support.

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